

DIFFERENTIAL GEOMETRY AND LIE GROUPS

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Fifth Semester

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Chapter 1

PRELIMINARIES AND MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds*. Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

1.1 Calculus on $\mathbb{R}^n$

A point p or element in $\mathbb{R}^n$ is associated with an n -tuple of coordinates $x^1, x^2, \dots, x^n$. Let U be an open neighbourhood of p in $\mathbb{R}^n$. For a non-negative integer k , we say that a function $f : U \rightarrow \mathbb{R}$ is C^k -continuous at $p \in U$ if all partial derivatives of f of order $\leq k$ exist and are continuous at p . f is said to be C^∞ or *smooth* if f is C^k for all $k \geq 0$. f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U .

If f happens to be a vector-valued function, i.e. $f : U \rightarrow \mathbb{R}^m$, we say $f = (f^1, f^2, \dots, f^m)$ is C^k (or C^∞) at p if each component function $f^i : U \rightarrow \mathbb{R}$ is C^k (or C^∞) at p . f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U .

Definition 1.1. Let R be a commutative ring with unity. We term A to be an R -algebra if A is a commutative ring with unity and an R -module, and for $r \in R$ and $a, b \in A$, we have $r(ab) = (ra)b = a(rb)$.

In this course, we will be primarily be concerned with the case when R is a field, and A is an R -vector space.

For a subset $U \subseteq \mathbb{R}^n$, we denote $C^k(U)$ (or $C^\infty(U)$) as the set of all C^k (or C^∞) functions from U to $\mathbb{R}$. Note that this is a commutative ring with unity. It is also a vector space over $\mathbb{R}$. Together, $C^k(U)$ (or $C^\infty(U)$) is a (commutative) algebra over $\mathbb{R}$. Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \dots \supseteq C^k(U) \supseteq C^{k+1}(U) \supseteq \dots \quad (1.1)$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \quad (1.2)$$

$f : U \rightarrow \mathbb{R}$ is termed *real analytic* at $p \in U$ if there exists a neighbourhood $V \subseteq U$ of p such that

$$f(x) = f(p) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \dots + \frac{1}{k!} \sum_{1 \leq i_1, \dots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \dots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \dots (x^{i_k} - p^{i_k}) + \dots \quad (1.3)$$

for all $x \in V$. f is said to be real analytic on U if it is real analytic at every point in U . We denote the set of all real analytic functions on U as $C^\omega(U)$. Note that $C^\omega(U) \subseteq C^\infty(U)$.

Theorem 1.2 (*Taylor's theorem*). *Let $p \in U \subseteq \mathbb{R}^n$, and $f : U \rightarrow \mathbb{R}$ be a C^∞ function. Further assume that for all $x \in U$, the line segment from p to x is contained in U (U in this case is said to be star-shaped with respect to p). Then there exist $g_1, \dots, g_n \in C^\infty(U)$ such that, for all $x \in U$,*

$$f(x) = f(p) + \sum_{i=1}^n g_i(x)(x^i - p^i), \quad (1.4)$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \leq i \leq n. \quad (1.5)$$

Proof. Let $x \in U$. Then the line segment $\{p + t(x - p) \mid t \in [0, 1]\}$ is contained in U . Composing f with the map $t \mapsto p + t(x - p)$, we obtain

$$\frac{df}{dt}(p + t(x - p)) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p + t(x - p))(x^i - p^i). \quad (1.6)$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x - p))dt = \sum_{i=1}^n (x^i - p^i) \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x - p))dt = \sum_{i=1}^n (x^i - p^i)g_i(x). \quad (1.7)$$

Clearly, because f is C^∞ , each g_i is also C^∞ . Further, we have $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$. ■

Let $p = (p^1, \dots, p^n)$, $v \in \mathbb{R}^n$, and $f : U \rightarrow \mathbb{R}$ be C^∞ at p . The *directional derivative* of f at p in the direction of v is defined as

$$\lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t} = \left. \frac{d}{dt} f(p + tv) \right|_{t=0} = \sum_{i=1}^n \frac{df}{dx^i}(p)v^i. \quad (1.8)$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p \quad (1.9)$$

Now consider all pairs of the form (f, U) where $p \in U \subseteq \mathbb{R}^n$ is an open neighbourhood of p and $f : U \rightarrow \mathbb{R}$ is C^∞ . Define an equivalence relation $\sim$ on this set by declaring $(f, U) \sim (g, V)$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) is denoted by $[(f, U)]$, and the set of all such equivalence classes is denoted by $C_p^\infty(\mathbb{R}^n)$. Such an equivalence class $[(f, U)]$ is called the *germ* of f at p . We can give $C_p^\infty(\mathbb{R}^n)$ the structure of a commutative ring with unity by defining

$$[(f, U)] + [(g, V)] = [(f + g, U \cap V)], \quad [(f, U)] \cdot [(g, V)] = [(fg, U \cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \quad (1.10)$$

It can also be given the structure of a vector space over $\mathbb{R}$ by defining

$$\lambda[(f, U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \quad (1.11)$$

Thus, $C_p^\infty(\mathbb{R}^n)$ is an $\mathbb{R}$ -algebra. Now, given a germ $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$ and a vector $v \in \mathbb{R}^n$, we define

$$D_v \alpha = \sum_{i=1}^n v^i \frac{\partial f}{\partial x^i} \Big|_p = v \cdot \nabla f(p). \quad (1.12)$$

Note that the choice of f is immaterial, since if $[(f, U)] = [(g, V)]$, then there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$, and hence $\nabla f(p) = \nabla g(p)$. One can infer that $D_{\lambda v}(f) = \lambda D_v(f) =$

$D_v(\lambda f)$ and $D_v(f + g) = D_v(f) + D_v(g)$ for all $\lambda \in \mathbb{R}$ and $f, g \in C_p^\infty(\mathbb{R}^n)$. Thus, D_v is an $\mathbb{R}$ -linear map from $C_p^\infty(\mathbb{R}^n)$ to $\mathbb{R}$ (hereforth, we shall use f to also denote the germ $[(f, U)]$ of f at p). Also,

$$D_v(fg) = \sum_{i=1}^n v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left(f(p) \frac{\partial g}{\partial x^i}(p) + g(p) \frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \quad (1.13)$$

Such a property makes D_v what we call a derivation.

Definition 1.3. Any $\mathbb{R}$ -linear map $T : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ satisfying the Leibniz rule $T(fg) = f(p)T(g) + g(p)T(f)$ for all $f, g \in C_p^\infty(\mathbb{R}^n)$ is called a *derivation* at p .

One may show that if T is a derivation at p , then $T(1) = 0$.

The set of all derivations at p is denoted by $\mathfrak{D}_p(\mathbb{R}^n)$. It is easy to see that $\mathfrak{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. Now consider the map

$$\theta : \mathbb{R}^n \rightarrow \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \quad (1.14)$$

This θ is an $\mathbb{R}$ -linear map (verify). We claim that θ is an isomorphism of real vector spaces. To this end, let $v \in \mathbb{R}^n$ such that $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$. Then $D_v f = 0$ for all $f \in C_p^\infty(\mathbb{R}^n)$. In particular, for the coordinate functions x^i , we have $D_v x^i = v_i = 0$ for all $1 \leq i \leq n$. Thus, $v = 0$, and hence θ is injective. To show surjectivity, we appeal to Taylor's theorem; let $D \in \mathfrak{D}_p(\mathbb{R}^n)$. We claim that $D = D_v$ where $v = (D(x^1), \dots, D(x^n))$. Indeed, for any $f \in C_p^\infty(\mathbb{R}^n)$, we have

$$\begin{aligned} D(f) &= D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i)g_i(x)\right) = 0 + \sum_{i=1}^n ((x^i - p^i)(p)Dg_i(x) + D(x^i - p^i)g_i(p)) \\ &= \sum_{i=1}^n (Dx^i)g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)Dx^i = D_v(f). \end{aligned} \quad (1.15)$$

Hence, θ is surjective, and we have the desired isomorphism $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$.

Note that a germ $f \in C_p^\infty(\mathbb{R}^n)$ has a valid value at p , but no other point. Hence, we have the evaluation map

$$\text{ev}_p : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad [(f, U)] \mapsto f(p). \quad (1.16)$$

The kernel here is $\{[(f, U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$, which is an ideal in $C_p^\infty(\mathbb{R}^n)$. For any other element $[(f, U)] \in C_p^\infty(\mathbb{R}^n)$ with $f(p) \neq 0$, a valid inverse is given by $[(f^{-1}, U \setminus \{f = 0\})]$. This makes $C_p^\infty(\mathbb{R}^n)$ a local ring.

1.2 Multilinear Algebra

July 22nd.

Hereforth, V shall denote a finite-dimensional real vector space, and V^V shall denote its *dual space* $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$. If $e_1, \dots, e_n$ are a basis for V , then the *dual basis* of V^V is denoted by $\alpha^1, \dots, \alpha^n \in V^V$, and is defined by $\alpha^i(e_j) = \delta_{i,j}$ for all $1 \leq i, j \leq n$. One may verify that the dual basis $\{\alpha^1, \dots, \alpha^n\}$, induced by the basis $\{e_1, \dots, e_n\}$, is indeed a basis for V^V . S_k shall denote the permutation group on k symbols, satisfying $|S_k| = k!$. For any $\sigma \in S_k$, the sign of σ is denoted by $\text{sgn}(\sigma)$, and is defined as $+1$ if σ is an even permutation, and -1 if σ is an odd permutation.

With notation out of the way, let $k \geq 1$ be an integer. We shall consider the k -tuple $V^k = V \times \dots \times V$ (k times).

Definition 1.4. A mapping $f : V^k \rightarrow \mathbb{R}$ is said to be *k-linear* if f is linear in each of its k arguments.

A special case is when $k = 2$, in which case f is called a *bilinear form*. Such an $f : V \times V \rightarrow \mathbb{R}$ satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v, u) + \mu f(w, u), \quad f(u, \lambda v + \mu w) = \lambda f(u, v) + \mu f(u, w). \quad (1.17)$$

A common example of a bilinear form is any valid inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ on V . Another example is the determinant $\det : \mathbb{R}^n \rightarrow \mathbb{R}$, where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \rightarrow \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \quad (1.18)$$

It is easy to see that $L_k(V)$ is an $\mathbb{R}$ -vector space.

Definition 1.5. A k -linear map $f : V^k \rightarrow \mathbb{R}$ is called a *symmetric k -linear form* if for all $\sigma \in S_k$,

$$f(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.19)$$

That is, f is invariant under any permutation of its arguments. Moreover, f is called an *alternating k -linear form* if for all $\sigma \in S_k$,

$$f(v_1, \dots, v_k) = \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.20)$$

One can see that the standard inner product on $\mathbb{R}^n$ is a symmetric bilinear form, and the determinant is an alternating n -linear form. Related to these definitions, we can immediately define the following subspaces of $L_k(V)$:

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \quad (1.21)$$

Currently, our main interest of study is $A_k(V)$, the space of alternating k -linear forms on V . Elements of $A_k(V)$ are termed *alternating k -tensors*. Similarly, those of $S_k(V)$ and $L_k(V)$ are called *symmetric k -tensors* and *k -tensors*, respectively. For $k = 0$, we define $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$. For $k = 1$, we have $L_1(V) = S_1(V) = A_1(V) = V^V$. We can define the (left) action of S_k on $L_k(V)$ by

$$(\sigma \cdot f)(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.22)$$

Thus, f is symmetric if $\sigma \cdot f = f$ for all $\sigma \in S_k$, and f is alternating if $\sigma \cdot f = \text{sgn}(\sigma)f$ for all $\sigma \in S_k$.

Given an $f \in L_k(V)$, we can define the *symmetrization* $Sf \in S_k(V)$, of f , as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \quad (1.23)$$

One may, indeed, check that Sf is symmetric. Similarly, we can define the *alternation* $Af \in A_k(V)$, of f , as

$$Af = \sum_{\sigma \in S_k} \text{sgn}(\sigma) \sigma \cdot f. \quad (1.24)$$

1.2.1 The Tensor and Wedge Products

July 24th.

For any $f \in L_k(V)$ and $g \in L_l(V)$, we can define the *tensor product* $f \otimes g \in L_{k+l}(V)$ as

$$(f \otimes g)(v_1, \dots, v_{k+l}) = f(v_1, \dots, v_k)g(v_{k+1}, \dots, v_{k+l}). \quad (1.25)$$

Verify that $f \otimes g$ is indeed $(k+l)$ -linear. We are interested in the tensor product of alternating tensors, so let $f \in A_k(V)$ and $g \in A_l(V)$. Then, we can define the *wedge product* $f \wedge g \in A_{k+l}(V)$ as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot (f \otimes g). \quad (1.26)$$

The associativity of the tensor product holds easily: for $f \in L_k(V)$, $g \in L_l(V)$, and $h \in L_m(V)$,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \quad (1.27)$$

Proposition 1.6. For $f \in A_k(V)$ and $g \in A_l(V)$, we have

$$f \wedge g = (-1)^{kl} g \wedge f. \quad (1.28)$$

Proof. We shall show that $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$. To this end, fix the permutation $\tau \in S_{k+l}$ defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \quad (1.29)$$

We have

$$\begin{aligned} A(f \otimes g)(v_1, \dots, v_{k+l}) &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}) \\ &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) f(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}) \\ &= \text{sgn}(\tau) \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}) \\ &= \text{sgn}(\tau) A(g \otimes f)(v_1, \dots, v_{k+l}). \end{aligned} \quad (1.30)$$

With $\text{sgn}(\tau) = (-1)^{kl}$, the proof is complete. $\blacksquare$

Corollary 1.7. For $f \in A_k(V)$, we have $f \wedge f = 0$ if k is odd.

Proof. Easy since $f \wedge f = (-1)^{k^2} f \wedge f = -f \wedge f$ if k is odd, so $f \wedge f = 0$. $\blacksquare$

Lemma 1.8. For $f \in L_k(V)$ and $g \in L_l(V)$, we have

$$A(Af \otimes g) = k! A(f \otimes g), \quad A(f \otimes Ag) = l! A(f \otimes g). \quad (1.31)$$

Proof. We shall prove the first equality; the second follows similarly. We have

$$\begin{aligned} A(Af \otimes g) &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \text{sgn}(\tau) (\tau \cdot f) \otimes g \right) = \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \text{sgn}(\tau) \tau \cdot (f \otimes g) \right) \\ &= \sum_{\tau \in S_k} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) \sigma\tau \cdot (f \otimes g) = k! \sum_{\alpha \in S_{k+l}} \text{sgn}(\alpha) \alpha \cdot (f \otimes g) = k! A(f \otimes g). \end{aligned} \quad (1.32)$$

Proposition 1.9. The wedge product is associative, i.e. for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$, we have

$$(f \wedge g) \wedge h = f \wedge (g \wedge h). \quad (1.33)$$

Proof. We have

$$\begin{aligned} (f \wedge g) \wedge h &= \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)!m!k!l!} A(A(f \otimes g) \otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!} A((f \otimes g) \otimes h) \\ &= \frac{1}{k!l!m!} A(f \otimes g \otimes h) = f \wedge (g \wedge h). \end{aligned} \quad (1.34)$$

Corollary 1.10. $f \wedge g \wedge h = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$ for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$. Inductively,

$$f_1 \wedge \cdots \wedge f_k = \frac{1}{k_1! \cdots k_k!} A(f_1 \otimes \cdots \otimes f_k), \quad (1.35)$$

where $f_i \in A_{k_i}(V)$ for $1 \leq i \leq k$.

Proposition 1.11. Let $\alpha^1, \dots, \alpha^n \in V^V = A_1(V) = L_1(V)$, and $v_1, \dots, v_k \in V$. Then

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) = \det(\alpha^i(v_j))_{1 \leq i, j \leq k}. \quad (1.36)$$

Proof. We have

$$\begin{aligned} (\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) (\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) \alpha^1(v_{\sigma(1)}) \alpha^2(v_{\sigma(2)}) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i, j \leq k}. \end{aligned} \quad (1.37)$$

■

July 27th.

Let V be an $\mathbb{R}$ -vector space with basis $e_1, \dots, e_n$, and V^V be its dual space with dual basis $\alpha^1, \dots, \alpha^n$. Moreover, let I be a multi-index of length k , i.e. $I = (i_1, \dots, i_k)$ with $1 \leq i_1, \dots, i_k \leq n$. We define $e_I := (e_{i_1}, \dots, e_{i_k}) \in V^k$ and $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$. Then, a k -linear form $f \in L_k(V)$ is completely determined by its values on the e_I , and an alternating k -linear form $f \in A_k(V)$ is completely determined by its values on the e_I with $i_1 < i_2 < \cdots < i_k$.

To this end, we first claim that $\alpha^I(e_J) = \delta_J^I$ for all multi-indices I, J of length k . Indeed, if $J = (j_1, \dots, j_k)$ with $j_1 < j_2 < \cdots < j_k$, then

$$\alpha^I(e_J) = \det(\alpha^{i_m}(e_{j_n}))_{1 \leq m, n \leq k}; \quad (1.38)$$

. this is 1 for $I = J$. Suppose $I \neq J$. Let $1 \leq l \leq k$ be the first index where $i_l \neq j_l$. Then the l^{th} row of $\alpha^{i_m}(e_{j_n})_{1 \leq m, n \leq k}$ is an all zeroes row, and hence $\det(\alpha^{i_m}(e_{j_n}))_{1 \leq m, n \leq k} = 0$. Thus, $\alpha^I(e_J) = \delta_J^I$. We now claim that α^I , for strictly increasing multi-indices I of length k , form a basis for $A_k(V)$. For linear independence, we simply have (the sums in this context are over strictly increasing multi-indices I of length k):

$$\sum_I c_I \alpha^I = 0 \implies \sum_I c_I \alpha^I(e_J) = 0 \implies c_J = 0, \quad \forall J. \quad (1.39)$$

To show that they span $A_k(V)$, let $f \in A_k(V)$. Then consider the k -linear form

$$g = \sum_I f(e_I) \alpha^I. \quad (1.40)$$

We show that $f = g$; it is enough to prove that $f(e_J) = g(e_J)$ for all strictly increasing multi-indices J of length k . We have

$$g(e_J) = \sum_I f(e_I) \alpha^I(e_J) = f(e_J) \alpha^J(e_J) + \sum_{I \neq J} f(e_I) \alpha^I(e_J) = f(e_J). \quad (1.41)$$

Corollary 1.12. $A_k(V) = 0$ if $k > n$. Further, $\dim A_k(V) = \binom{n}{k}$ if $1 \leq k \leq n$.

1.3 Building Blocks for Manifolds

Recall that we had defined the class of germs $C_p^\infty(\mathbb{R}^n)$ of C^∞ functions at a point $p \in \mathbb{R}^n$, which agree on some neighbourhood of p . One can show that $C_p^\infty(\mathbb{R}^n)$ and $C_p^\infty(U)$ are isomorphic as $\mathbb{R}$ -algebras, where $U \subseteq \mathbb{R}^n$ is an open neighbourhood of p . Note that we term $T_p(U) = T_p(\mathbb{R}^n) = \mathbb{R}^n$ as the *tangent space* of U (or $\mathbb{R}$) at p .

Definition 1.13. Let $U \subseteq \mathbb{R}^n$ be an open subset. A *vector field* on U is a function $X : U \rightarrow \coprod_{p \in U} T_p(\mathbb{R}^n)$ such that $X(p) \in T_p(\mathbb{R}^n)$ for all $p \in U$. In particular,

$$X(p) = \sum_{i=1}^n a^i(p) \frac{\partial}{\partial x^i} \Big|_p, \quad a^i(p) \in \mathbb{R}. \quad (1.42)$$

We can rewrite this as

$$X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}, \quad a^i : U \rightarrow \mathbb{R}. \quad (1.43)$$

X is called a *smooth vector field* if each a^i is a smooth function on U . The set of all smooth vector fields on U is denoted by $\mathfrak{X}(U)$.

This is a real vector space; note that for $X, Y \in \mathfrak{X}(U)$, we have $X + Y \in \mathfrak{X}(U)$, and for $\lambda \in \mathbb{R}$, we have $\lambda X \in \mathfrak{X}(U)$. Further, $\mathfrak{X}(U)$ is a $C^\infty(U)$ -module since for $f \in C^\infty(U)$ and $X \in \mathfrak{X}(U)$, we have $fX \in \mathfrak{X}(U)$, where

$$(fX)(p) = f(p)X(p), \quad p \in U. \quad (1.44)$$

Notice that $X \in \mathfrak{X}(U)$ describes a map $C^\infty(U) \rightarrow C^\infty(U)$, where $f \mapsto Xf$, being defined as $(Xf)(p) = X_p f$. This is a derivation on $C^\infty(U)$, since for $f, g \in C^\infty(U)$, we have

$$(X(fg))(p) = X_p(fg) = f(p)X_p g + g(p)X_p f = (fXg)(p) + (gXf)(p). \quad (1.45)$$

We refine our definition of a derivation as follows.

Definition 1.14. Let K be a field, and A be a K -algebra. A *derivation* $D : A \rightarrow A$ is an $\mathbb{R}$ -linear map satisfying the Leibniz rule $D(ab) = aD(b) + bD(a)$ for all $a, b \in A$.

Definition 1.15. A *differential 1-form* on U is a smooth function $\omega : U \rightarrow \coprod_{p \in U} T_p(\mathbb{R}^n)^V$ such that $\omega(p) \in T_p(\mathbb{R}^n)^V =: T_p^*(\mathbb{R}^n)$ for all $p \in U$.

For example, for $f \in C^\infty(U)$, we can define the differential 1-form df on U by $df(p) = df_p \in \text{Hom}_{\mathbb{R}}(T_p \mathbb{R}^n, \mathbb{R})$, where $df_p(X) = X_p f$ for all $X \in T_p \mathbb{R}^n$. In particular, dx^i is a differential 1-form on U for all $1 \leq i \leq n$. In fact, $\{dx^1, \dots, dx^n\}$ is the dual basis of $T_p^* \mathbb{R}^n$ corresponding to the basis $\{\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p\}$ of $T_p \mathbb{R}^n$. Thus, a differential 1-form ω on U can be expressed as

$$\omega(p) = \sum_{i=1}^n a_i(p) dx^i \Big|_p, \quad a_i : U \rightarrow \mathbb{R}. \quad (1.46)$$

$T_p^*(\mathbb{R}^n)$, here, is termed the *cotangent space* of $\mathbb{R}^n$ at p . Thus, our current setup is as follows: $V = T_p \mathbb{R}^n$, and $V^V = T_p^* \mathbb{R}^n$. The basis for $T_p \mathbb{R}^n$ considered is $\{\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p\}$, and the corresponding dual basis for $T_p^* \mathbb{R}^n$ is $\{dx^1 \Big|_p, \dots, dx^n \Big|_p\}$. And finally, we have

$$A_k(T_p \mathbb{R}^n) = \text{the space of alternating } k\text{-linear forms on } T_p \mathbb{R}^n \quad (1.47)$$

which has the basis $\{dx^{i_1} \Big|_p \wedge \dots \wedge dx^{i_k} \Big|_p \mid 1 \leq i_1 < \dots < i_k \leq n\}$.

Definition 1.16. A *differential k -form* on U is a smooth function $\omega : U \rightarrow \coprod_{p \in U} A_k(T_p \mathbb{R}^n)$ such that $\omega(p) \in A_k(T_p \mathbb{R}^n)$ for all $p \in U$.

Any such ω may be written as $\omega = \sum_I a_I dx_I$. It termed a smooth differential k -form if each a_I is a smooth function on U . The set of all smooth differential 1-forms on U is denoted by $\Omega^1(U)$, and the set

of all smooth differential k -forms on U is denoted by $\Omega^k(U)$. Both these sets are $\mathbb{R}$ -vector spaces, and $\Omega^k(U)$ is a free $C^\infty(U)$ -module of rank $\binom{n}{k}$. The isomorphism may be given as

$$\Omega^1(U) \cong C^\infty(U)^{\oplus \binom{n}{1}}, \quad \omega = \sum_I a_I dx_I \mapsto (a_1, a_2, \dots, a_{\binom{n}{1}}). \quad (1.48)$$

We can endow $\Omega^k(U)$ with the wedge product $\wedge : \Omega^k(U) \times \Omega^l(U) \rightarrow \Omega^{k+l}(U)$, defined as

$$(\omega \wedge \eta)(p) = \omega(p) \wedge \eta(p), \quad p \in U, \quad (1.49)$$

or

$$\omega \wedge \eta = \sum_{I, J} a_I b_J dx_I \wedge dx_J \quad (1.50)$$

where I and J are strictly increasing multi-indices of length k and l , respectively. Note that if $I \cap J \neq \emptyset$, then $dx_I \wedge dx_J = 0$. Thus, the sum may be taken over all strictly increasing multi-indices I and J of length k and l , respectively, such that $I \cap J = \emptyset$. Also,

$$A_*(V) = \bigoplus_{k=0}^n A_k(V) \quad (1.51)$$

is defined as the *exterior algebra* of V . The wedge product on here is associative and anti-commutative. We define $\Omega^0(U) = C^\infty(U)$. The map $d : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$ is defined as the *exterior derivative*; for $k = 0$, we have

$$d : C^\infty(U) \rightarrow \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i. \quad (1.52)$$

For a k -form $\omega = \sum_I a_I dx_I \in \Omega^k(U)$, we have

$$d\omega = \sum_I da_I \wedge dx_I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx_I. \quad (1.53)$$

Example 1.17. Let $U \subseteq \mathbb{R}^3$. Let us call the coordinates $x^1 = x$, $x^2 = y$, and $x^3 = z$. The only non-zero forms are $\Omega^0(U)$, $\Omega^1(U)$, $\Omega^2(U)$, and $\Omega^3(U)$. Note that $\Omega^0(U) = C^\infty(U)$, and there's nothing to say about this. For $\Omega^1(U)$, a form looks like $f dx + g dy + h dz$ for $f, g, h \in C^\infty(U)$. For $\Omega^2(U)$, a form looks like $R dx \wedge dy + Q dx \wedge dz + P dy \wedge dz$ for $f, g, h \in C^\infty(U)$.

July 29th.

The map $\Omega^1(U) \times \mathfrak{X}(U) \rightarrow C^\infty(U)$, defined as $(\omega, X) \mapsto \omega(X)$, is $\mathbb{R}$ -bilinear. In particular, for $\omega = \sum_{i=1}^n a_i dx^i$ and $X = \sum_{j=1}^n b^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X) = \sum_{i=1}^n a_i dx^i \left(\sum_{j=1}^n b^j \frac{\partial}{\partial x^j} \right) = \sum_{i=1}^n a_i b^i \in C^\infty(U). \quad (1.54)$$

Extending this, we have the map $\Omega^k(U) \times \mathfrak{X}(U)^k \rightarrow C^\infty(U)$, defined as $(\omega, X_1, \dots, X_k) \mapsto \omega(X_1, \dots, X_k)$, which is $\mathbb{R}$ -multilinear. In particular, for $\omega = \sum_I b_I dx^I$ and $X_i = \sum_{j=1}^n a_i^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X_1, \dots, X_k) = \sum_I b_I a_{i_1}^1 \cdots a_{i_k}^k \in C^\infty(U) \quad (1.55)$$

using the fact that $dx^I(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}}) = 1$ if the multi-index matches, and 0 otherwise.

1.3.1 Exterior Differentiation

Here, d_k shall denote the map $d_k : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$, defined as the *exterior derivative*. The subscript k is often omitted, and we simply write $d : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$. For $k = 0$, we have

$$d : C^\infty(U) \rightarrow \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i. \quad (1.56)$$

Analogously, for a general k -form $\omega = \sum_I a_I dx^I \in \Omega^k(U)$, we have

$$d : \Omega^k(U) \rightarrow \Omega^{k+1}(U), \quad \omega \mapsto d\omega = \sum_I da_I \wedge dx^I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx^I. \quad (1.57)$$

All the d maps are $\mathbb{R}$ -linear, but not $C^\infty(U)$ -linear. Also, for two forms $\omega \in \Omega^k(U)$ and $\eta \in \Omega^l(U)$, we have

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta. \quad (1.58)$$

This is known as *antiderivation*. Moreover, $d_{k+1} \circ d_k = d^2 = 0$. Both these properties may be verified by the reader. Now suppose we have another map $D : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$ which is $\mathbb{R}$ -linear, satisfies antiderivation, and $D^2 = 0$, and $Df = df$ for all $f \in C^\infty(U)$. Then $D = d$; the exterior derivative is unique. Thus, a chain of maps is formed:

$$0 \xrightarrow{d} C^\infty(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(U) \xrightarrow{d} 0 \quad \text{for open } U \subseteq \mathbb{R}^n. \quad (1.59)$$

An $\omega \in \Omega^k(U)$ is called an *exact k -form* if $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(U)$. It is called a *closed k -form* if $d\omega = 0$. Note that every exact form is closed, but the converse is not true in general. We will refine the following later: quotienting the space of closed k -forms by the space of exact k -forms gives us the *de Rham cohomology* of U , denoted by $H_{\text{dR}}^k(U)$.

1.4 Manifolds

July 31st.

Definition 1.18. A topological space M is termed a *topological manifold* of dimension n if

1. M is Hausdorff,
2. M is second countable (that is, has a countable basis), and
3. for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ and a homeomorphism $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$, where $\varphi(U)$ is open. This makes a manifold *locally euclidean*.

The *invariance of dimension* property states that if we have two open sets $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ such that U is homeomorphic to V , then $m = n$. Thus, the dimension of a topological manifold is well-defined.

Definition 1.19. A pair (U, φ) is called a *coordinate chart* on M if $U \subseteq M$ is open and $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.

A trivial example is $\mathbb{R}^n$ with one (coordinate) chart $(\mathbb{R}^n, \text{id}_{\mathbb{R}^n})$. In fact, if $U \subseteq \mathbb{R}^n$ is open, then (U, id_U) is a coordinate chart. Note that subspaces inherit the properties of Hausdorffness and second countability. We state some properties without proof; if X is locally Euclidean but not T_2 , then X being a topological manifold implies that the number of connected (or path connected) components is at most countable.

Definition 1.20. Two charts (U, φ) and (V, ψ) on a topological manifold M are said to be *C^∞ -compatible* if either

1. $U \cap V = \emptyset$, or
2. $U \cap V \neq \emptyset$, and the transition maps $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$ and $\varphi \circ \psi^{-1} : \psi(U \cap V) \rightarrow \varphi(U \cap V)$ are C^∞ -maps between open subsets of $\mathbb{R}^n$.

Similarly, C^k -compatibility may be defined for $k \geq 0$. A C^ω -manifold is termed to be a real analytic manifold. In case of $k = 2m$, then $\mathbb{R}^{2m}$ is identified with $\mathbb{C}^m$, and the transition maps are required to be holomorphic.

Definition 1.21. M is a C^∞ -manifold or a *smooth manifold* if M is a topological manifold and there exists a family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that

1. $\{U_\alpha\}_{\alpha \in \Lambda}$ is an open cover of M , and
2. $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are C^∞ -compatible for all $\alpha, \beta \in \Lambda$.

Similarly, C^k -manifolds, C^ω -manifolds, and complex manifolds may be defined. The family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ is called an *atlas* of M .

The most basic non-trivial example is perhaps the circle $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$, which is a C^∞ -manifold of dimension 1. We can think of $\mathbb{R}^2$ as $\mathbb{C}$. Let $U = S^1 \setminus \{1\} = \{e^{it} \mid 0 < t < 2\pi\}$ and $V = S^1 \setminus \{-1\} = \{e^{it} \mid -\pi < t < \pi\}$. Then (U, φ) and (V, ψ) are charts, where $\varphi : U \rightarrow (0, 2\pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$ and $\psi : V \rightarrow (-\pi, \pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$. Here, $U \cap V = \{e^{it} \mid 0 < t < \pi\} \cup \{e^{-t} \mid \pi < t < 2\pi, -\pi < t < 0\}$. The transition maps are $\psi \circ \varphi^{-1} : (0, \pi) \rightarrow (0, \pi)$ defined as $t \mapsto t$ and $\psi \circ \varphi^{-1} : (\pi, 2\pi) \rightarrow (-\pi, 0)$ defined as $t \mapsto t - 2\pi$. Both these maps are smooth, and hence the charts are compatible. Thus, S^1 is a smooth manifold of dimension 1.

August 3rd.

Proposition 1.22. Let $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ be an atlas on M . Suppose (V, ψ) and (W, σ) are two charts on M which are C^∞ -compatible with all charts in $\mathcal{U}$. Then (V, ψ) and (W, σ) are C^∞ -compatible.

Proof. Let $p \in V \cap W$. Then there exists $\alpha \in \Lambda$ such that $p \in U_\alpha$. Since (V, ψ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\psi \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap V) \rightarrow \psi(U_\alpha \cap V)$ is smooth. Similarly, since (W, σ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\sigma \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap W) \rightarrow \sigma(U_\alpha \cap W)$ is smooth. Now, we have

$$(\sigma \circ \psi^{-1})|_{\psi(U_\alpha \cap V \cap W)} = (\sigma \circ \varphi_\alpha^{-1})|_{\varphi_\alpha(U_\alpha \cap V \cap W)} \circ (\varphi_\alpha \circ \psi^{-1})|_{\psi(U_\alpha \cap V \cap W)}. \quad (1.60)$$

The right-hand side is a composition of smooth maps, and hence is smooth. Thus, $\sigma \circ \psi^{-1}$ is smooth on an open neighbourhood of p . Since p was arbitrary, we have that $\sigma \circ \psi^{-1}$ is smooth on $V \cap W$. Similarly, we can show that $\psi \circ \sigma^{-1}$ is smooth on $V \cap W$. Thus, (V, ψ) and (W, σ) are compatible. ■

This leads to the following definition.

Definition 1.23. A *maximal atlas* $\mathcal{M}$ on a smooth manifold M is an atlas which is maximal with respect to inclusion; that is, if $\mathcal{M} \subseteq \mathcal{U}$ for some atlas $\mathcal{U}$, then $\mathcal{M} = \mathcal{U}$. Equivalently, $\mathcal{M}$ is a maximal atlas if it contains all charts which are compatible with all charts in $\mathcal{M}$.

Any atlas $\mathcal{U}$ on M is contained in a unique maximal atlas $\mathcal{M}$. For a proof of this, let $\mathcal{M} = \{(U, \varphi) \mid (U, \varphi) \text{ compatible with all charts in } \mathcal{U}\}$. Then $\mathcal{M}$ is an atlas. If $\mathcal{M} \subseteq \mathcal{M}'$, then any chart in $\mathcal{M}'$ must be compatible with all charts in $\mathcal{M}$; it must be compatible with all charts in $\mathcal{U}$. By definition of $\mathcal{M}$, such a chart must be in $\mathcal{M}$, so $\mathcal{M} = \mathcal{M}'$. For uniqueness, if $\mathcal{U} \subseteq \mathcal{M}, \mathcal{M}'$, then $\mathcal{M}, \mathcal{M}' \subseteq \mathcal{M} \cup \mathcal{M}'$ tells us $\mathcal{M} = \mathcal{M}'$ by maximality. The unique maximal atlas $\mathcal{M} \supseteq \mathcal{U}$ is also called a differentiable structure on M .

Example 1.24. 1. Let $U \subseteq \mathbb{R}^n$, and $F : U \rightarrow \mathbb{R}^m$ be a C^∞ -map. The graph of F is defined as $\Gamma(F) = \{(x, F(x)) \mid x \in U\} \subseteq \mathbb{R}^{n+m}$. An atlas on $\Gamma(F)$ is given by $\{(U, \varphi)\}$, where $\varphi : U \rightarrow \Gamma(F)$ is defined as $x \mapsto (x, F(x))$. The transition map is $\varphi^{-1} : \Gamma(F) \rightarrow U$ defined as $(x, F(x)) \mapsto x$. This is smooth, and hence $\Gamma(F)$ is a smooth manifold of dimension n .

2. Let M and N be smooth manifolds of dimension m and n , respectively. Then $M \times N$ is a smooth manifold of dimension $m + n$; if $\mathcal{U}$ and $\mathcal{V}$ are atlases on M and N , respectively, then $\mathcal{U} \times \mathcal{V} = \{(U \times V, \varphi \times \psi) \mid (U, \varphi) \in \mathcal{U}, (V, \psi) \in \mathcal{V}\}$ is an atlas on $M \times N$. The transition maps are $(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1} = (\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$, which is smooth since both $\varphi_2 \circ \varphi_1^{-1}$ and $\psi_2 \circ \psi_1^{-1}$ are smooth.

1.4.1 Smooth Functions and Smooth Maps

We will often omit ‘smooth’ but still mean smooth in the context of manifolds.

Definition 1.25. Let M be a manifold of dimension n . A function $f : M \rightarrow \mathbb{R}$ is smooth at $p \in M$ if there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$ is smooth at $\varphi(p)$. A function $f : M \rightarrow \mathbb{R}$ is smooth if it is smooth at all $p \in M$.

The definition of the smooth function $f : M \rightarrow \mathbb{R}$ is independent of the choice of chart (U, φ) , since if (V, ψ) is another chart containing p , then $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is smooth, and hence $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$ is smooth at $\psi(p)$.

Proposition 1.26. Let M be a manifold of dimension n , and $f : M \rightarrow \mathbb{R}$ be a function. Then the following are equivalent:

1. f is a smooth function on M .
2. M has an atlas $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that $f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha) \rightarrow \mathbb{R}$ is smooth for all $\alpha \in \Lambda$.
3. For all charts (U, φ) on M , the map $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$ is smooth.

Proof. Note that 2. implies 1. by definition, and 3. implies 2. trivially. We show that 1. implies 3. Let (U, φ) be a chart on M , and $p \in U$; it is enough to prove that $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. Since f is smooth, there exists a chart (V, ψ) of M such that $p \in V$ and $f \circ \psi^{-1} : \psi(V) \rightarrow \mathbb{R}$ is smooth at $\psi(p)$. Now, we have

$$f \circ \varphi^{-1} = (f \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1}). \quad (1.61)$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. ■

A function $f : M \rightarrow \mathbb{R}$ is smooth on a chart (U, φ) if and only if the pullback $\varphi^* f := f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$ is smooth. Also note that the definition of a smooth function f does not assume that f is continuous; however, one may show that the definition itself implies that f is continuous. This is because if f is smooth at $p \in M$, then there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$ is smooth at $\varphi(p)$, and hence continuous at $\varphi(p)$. Since φ is a homeomorphism, we have that f is continuous at p . Since p was arbitrary, we have that f is continuous on M .

Definition 1.27. Let M and N be smooth manifolds of dimension m and n , respectively. A continuous map $F : N \rightarrow M$ is smooth at $p \in N$ if there exist charts (U, φ) on N , with $p \in U$, and (V, ψ) on M , with $F(p) \in V$, such that $\psi \circ F \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V)$ is smooth at $\varphi(p)$. A continuous map $F : N \rightarrow M$ is smooth if it is smooth at all $p \in N$.

The definition of a map f being smooth at p is independent of the choice of charts (U, φ) and (V, ψ) , since if (U', φ') and (V', ψ') are other charts containing p and $F(p)$, respectively, then $\psi' \circ \psi^{-1} : \psi(V \cap V') \rightarrow \psi'(V \cap V')$ and $\varphi' \circ \varphi^{-1} : \varphi(U \cap U') \rightarrow \varphi'(U \cap U')$ are smooth, and hence

$$\psi' \circ F \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1}) \quad (1.62)$$

Proposition 1.28. Let $F : N \rightarrow M$ be a continuous map between smooth manifolds. Then the following are equivalent:

1. F is smooth.
2. There exist atlases $\mathcal{U}$ of N and $\mathcal{V}$ of M such that for all $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \rightarrow \psi(V)$ is smooth.
3. For all charts (U, φ) on N and (V, ψ) on M , the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \rightarrow \psi(V)$ is smooth.

Proof. Again, note that 2. implies 1. by definition, and 3. implies 2. trivially. A similar argument as in the previous proposition shows that 1. implies 3. ■

Proposition 1.29. *Let $F : N \rightarrow M$ and $G : M \rightarrow P$ be smooth maps between smooth manifolds. Then the composition $G \circ F : N \rightarrow P$ is smooth.*

Proof. Let (U, φ) , (V, ψ) , and (W, σ) be charts on N , M , and P , respectively, such that $p \in U$, $F(p) \in V$, and $G(F(p)) \in W$. Then we have

$$\sigma \circ (G \circ F) \circ \varphi^{-1} = (\sigma \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}). \quad (1.63)$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $G \circ F$ is smooth at p . ■

Finally, a smooth map $F : N \rightarrow M$ is termed a *diffeomorphism* if F is a bijection, and F and F^{-1} are smooth. Some properties of diffeomorphisms may be inferred.

Proposition 1.30. *The following hold true.*

1. *Let (U, φ) be a chart on a smooth manifold M of dimension n . Then $\varphi : U \rightarrow \varphi(U)$ is a diffeomorphism.*
2. *Suppose $U \subseteq M$ is open, and that $F : U \rightarrow F(U)$ is a diffeomorphism. Then (U, F) is a chart in the maximal atlas of M .*

Proof. 1. Note that U is a manifold since it is open in M , and $\varphi(U)$ is a manifold since it is open in $\mathbb{R}^n$. The map $\varphi : U \rightarrow \varphi(U)$ is a homeomorphism by definition of a chart, so it is bijective. Note that $\{(U, \varphi)\}$ is an atlas on U , and $\{(\varphi(U), \text{id}_{\varphi(U)})\}$ is an atlas on $\varphi(U)$. Then φ is smooth since $\text{id}_{\varphi(U)} \circ \varphi \circ \varphi^{-1} = \text{id}_{\varphi(U)}$ is smooth, and φ^{-1} is smooth since $\varphi \circ \varphi^{-1} \circ \text{id}_{\varphi(U)} = \text{id}_{\varphi(U)}$ is smooth. Thus, φ is a diffeomorphism.

2. Since U is open in M , it is a smooth manifold. Because $F : U \rightarrow F(U)$ is a diffeomorphism, it is in particular a homeomorphism. Hence F is bijective, and its image $F(U)$ is an open subset of $\mathbb{R}^n$ (as required for a chart). Let (V, ψ) be any chart on M with $U \cap V \neq \emptyset$. Then the transition maps

$$\psi \circ F^{-1} : F(U \cap V) \rightarrow \psi(V) \quad \text{and} \quad F \circ \psi^{-1} : \psi(U \cap V) \rightarrow F(U \cap V) \quad (1.64)$$

are smooth, since they are compositions of smooth maps. Therefore (U, F) is compatible with every chart in the maximal atlas of M , and so (U, F) is a chart in that maximal atlas. ■

Proposition 1.31. *Let N be a smooth manifold of dimension n , and $F : N \rightarrow \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*
2. *N has an atlas $\mathcal{U}$ such that for all $(U, \varphi) \in \mathcal{U}$, the map $F \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}^m$ is smooth.*
3. *For all charts (U, φ) on N , the map $F \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}^m$ is smooth.*

Proof. 3. implies 2. is obvious. For 2. implies 1., note that there exist atlases $\mathcal{U}$ and $\mathcal{V}$ of N and M respectively such that $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$ satisfy the smoothness condition. Since $\mathbb{R}^m$ has a global chart $(\mathbb{R}^m, \text{id}_{\mathbb{R}^m})$, we can take $\mathcal{V} = \{(\mathbb{R}^m, \text{id}_{\mathbb{R}^m})\}$. Then for any $(U, \varphi) \in \mathcal{U}$, we have $\text{id}_{\mathbb{R}^m} \circ F \circ \varphi^{-1} = F \circ \varphi^{-1}$ is smooth, which implies that F is smooth. For 1. implies 3., F is smooth and φ is a diffeomorphism; thus, φ^{-1} is smooth, and $F \circ \varphi^{-1}$ is smooth. ■

Proposition 1.32. *Let $F : N \rightarrow M$ be a continuous map between smooth manifolds. Then the following are equivalent.*

1. *F is smooth.*

2. M has an atlas $\mathcal{V}$ such that for all $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F : F^{-1}(V) \rightarrow \psi(V)$ is smooth.
3. For any chart (V, ψ) of M , the map $\psi \circ F : F^{-1}(V) \rightarrow \psi(V)$ is smooth.

The proof follows a similar structure to the one above.

Proposition 1.33. *Let $F : N \rightarrow \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. F is smooth.
2. $F^1, \dots, F^m : N \rightarrow \mathbb{R}$ are smooth, where $F = (F^1, \dots, F^m)$.

Examples

August 5th.

We look at some examples of smooth manifolds and smooth maps.

Example 1.34. For $S^n \subseteq \mathbb{R}^{n+1}$, the n -sphere, define $U_i^+ = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i > 0\}$ and $U_i^- = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i < 0\}$ for $i = 1, \dots, n+1$. Then $\{(U_i^+, \varphi_i^+), (U_i^-, \varphi_i^-)\}_{i=1}^{n+1}$ is an atlas on S^n , where $\varphi_i^+ : U_i^+ \rightarrow \mathbb{R}^n$ is defined as $(x_1, \dots, x_{n+1}) \mapsto (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1})$, and $\varphi_i^- : U_i^- \rightarrow \mathbb{R}^n$ is defined similarly. The transition maps are smooth, and hence S^n is a smooth manifold of dimension n .

Example 1.35. Recall $\mathbb{R}P^n$, the *real projective space* of dimension n , defined as the quotient $\mathbb{R}^{n+1} \setminus \{0\} / \sim$, where $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $(x_1, \dots, x_n) = (\lambda y_1, \dots, \lambda y_n)$. Let $(a_1 : a_2 : \dots : a_{n+1})$ denote the equivalence class of $(a_1, \dots, a_{n+1}) \in \mathbb{R}^{n+1} \setminus \{0\}$. This is a smooth manifold of dimension n ; an atlas is given by $\{(U_i, \varphi_i)\}_{i=1}^{n+1}$, where $U_i = \{(x_1 : \dots : x_{n+1}) \in \mathbb{R}P^n \mid x_i \neq 0\}$ and $\varphi_i : U_i \rightarrow \mathbb{R}^n$ is defined as

$$\varphi_i(x_1 : \dots : x_{n+1}) = \left(\frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_{n+1}}{x_i} \right). \quad (1.65)$$

Verify that the transition maps $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j)$ are smooth for all $i, j = 1, \dots, n+1$. We can also provide a smooth map $F : S^n \rightarrow \mathbb{R}P^n$ defined as $(x_1, \dots, x_{n+1}) \mapsto (x_1 : \dots : x_{n+1})$. This is a smooth map, and it is a 2-to-1 covering map.

Example 1.36. Similar to the above, we can define $\mathbb{P}_k^n$ to be the projective space of dimension n over the field k ; that is, $\mathbb{P}_k^n = k^{n+1} \setminus \{0\} / \sim$, where $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in k \setminus \{0\}$ such that $(x_1, \dots, x_{n+1}) = (\lambda y_1, \dots, \lambda y_{n+1})$. In particular, if $k = \mathbb{C}$, then $\mathbb{P}_{\mathbb{C}}^n$ consists of complex 1-dimensional subspaces of $\mathbb{C}^{n+1}$. This is a smooth manifold of dimension $2n$. $\mathbb{R}P^n$ may be, thus, interpreted as the space of real 1-dimensional subspaces of $\mathbb{R}^{n+1}$. Building upon this, the *Grassmannian space* $\text{Gr}(n, k)$ is defined as the space of k -dimensional subspaces of $\mathbb{R}^n$. This is a smooth manifold of dimension $k(n-k)$. The Grassmannian space $\text{Gr}(n+1, 1)$ is precisely $\mathbb{R}P^n$, and $\text{Gr}(n, 0)$ is a single point. $\text{Gr}(n, n-1)$ is the space of hyperplanes in $\mathbb{R}^n$. Again, the Grassmannian construction may be generalized to any field.

1.4.2 Quotient Manifolds

August 10th.

Recall that for an equivalence relation $\sim$ on a set S , the quotient set $S/\sim$ is defined as the set of equivalence classes $[s] = \{t \in S \mid t \sim s\} \in S/\sim$. If S happens to be a topological space, then the projection map $\pi : S \rightarrow S/\sim$ defined as $s \mapsto [s]$ is continuous if we endow $S/\sim$ with the quotient topology; that is, a subset $U \subseteq S/\sim$ is open if and only if $\pi^{-1}(U) \subseteq S$ is open. This is the finest topology on $S/\sim$ which makes π continuous. Also recall the universal property; that is, if $f : S \rightarrow X$ is a set map between topological spaces, and $\pi : S \rightarrow S/\sim$ is the projection map, then there exists a unique map $\tilde{f} : S/\sim \rightarrow X$

such that $f = \tilde{f} \circ \pi$ if and only if f is constant on equivalence classes. In this case, $\tilde{f}$ is continuous if and only if f is continuous.

The equivalence relation $\sim$ on S is called open if the projection map $\pi : S \rightarrow S/\sim$ is open; that is, if $U \subseteq S$ is open, then $\pi(U) \subseteq S/\sim$ is open. Usually, $S/\sim$ is not T_2 is general. The following is a sufficient condition for $S/\sim$ to be T_2 , or Hausdorff.

Proposition 1.37. *Let $\sim$ be an open equivalence relation on a topological space S . Then $S/\sim$ is T_2 if and only if $\{(x, y) \in S \times S \mid x \sim y\}$ is closed in $S \times S$.*

Proof. Suppose the set $R = \{(x, y) \in S \times S \mid x \sim y\}$ is closed in $S \times S$. That is, for all $(x, y) \in S \times S$ such that $x \not\sim y$, there exist open neighbourhoods U of x and V of y such that $U \times V \subseteq S \times S \setminus R$. Since $\sim$ is open, we have that $\pi(U)$ and $\pi(V)$ are open neighbourhoods of $[x]$ and $[y]$, respectively. We claim that $\pi(U) \cap \pi(V) = \emptyset$. Suppose not; then there exists $[z] \in \pi(U) \cap \pi(V)$. Then there exist $u \in U$ and $v \in V$ such that $[z] = [u] = [v]$, which implies that $u \sim v$. But this contradicts the fact that $(u, v) \in U \times V \subseteq S \times S \setminus R$. Thus, $\pi(U) \cap \pi(V) = \emptyset$, and hence $S/\sim$ is Hausdorff.

On the other hand, suppose $S/\sim$ is Hausdorff. Let $(x, y) \in S \times S$ such that $x \not\sim y$. Then there exist open neighbourhoods U of $[x]$ and V of $[y]$ in $S/\sim$ such that $U \cap V = \emptyset$. Since $\pi : S \rightarrow S/\sim$ is continuous, we have that $\pi^{-1}(U)$ and $\pi^{-1}(V)$ are open neighbourhoods of x and y , respectively. We claim that $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$. Suppose not; then there exists $(u, v) \in \pi^{-1}(U) \times \pi^{-1}(V)$ such that $u \sim v$. Then we have $[u] = [v]$, which implies that $[u] \in U \cap V$, contradicting the fact that $U \cap V = \emptyset$. Thus, $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$, and hence R is closed in $S \times S$. ■

Example 1.38. Let $\mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\}$ be the punctured space, and define an equivalence relation $\sim$ on it as $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $(x_1, \dots, x_{n+1}) = (\lambda y_1, \dots, \lambda y_{n+1})$. Then $\mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\} / \sim$ is precisely $\mathbb{R}P^n$. The projection map $\pi : \mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\} \rightarrow \mathbb{R}P^n$ is open; let $U \subseteq \mathbb{R}^n \setminus \{(0, \dots, 0)\}$ be open. Then $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \mathbb{R} \setminus \{0\}} \lambda U$ is open, and hence $\pi(U)$ is open in $\mathbb{R}P^n$. Also, the set $R = \{(x, y) \in (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \mid x \sim y\}$ is closed in $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$, since if $y_i = \lambda x_i$ for some $\lambda \in \mathbb{R} \setminus \{0\}$, then $x_i y_j - x_j y_i = 0$ for all $i, j = 1, \dots, n+1$. Thus, R is the zero set of the continuous functions $f_{ij} : (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \rightarrow \mathbb{R}$ defined as $(x, y) \mapsto x_i y_j - x_j y_i$ for all $i, j = 1, \dots, n+1$. Hence, R is closed in $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$. By the previous proposition, we have that $\mathbb{R}P^n$ is Hausdorff.

Proposition 1.39. *Let $\sim$ be an open equivalence relation on a topological space S . Then S is second countable if and only if $S/\sim$ is second countable.*

Proof. Suppose S is second countable, and let $\mathcal{B}$ be a countable basis for S . Then $\mathcal{B}' = \{\pi(B) \mid B \in \mathcal{B}\}$ is a countable collection of open sets in $S/\sim$. We claim that $\mathcal{B}'$ is a basis for $S/\sim$. Let $U \subseteq S/\sim$ be open, and let $[x] \in U$. Then $\pi^{-1}(U) \subseteq S$ is open, and hence there exists $B \in \mathcal{B}$ such that $x \in B \subseteq \pi^{-1}(U)$. Then we have $[x] \in \pi(B) \subseteq U$, and hence $\mathcal{B}'$ is a basis for $S/\sim$. Thus, $S/\sim$ is second countable.

On the other hand, suppose $S/\sim$ is second countable, and let $\mathcal{B}'$ be a countable basis for $S/\sim$. Then $\mathcal{B} = \{\pi^{-1}(B') \mid B' \in \mathcal{B}'\}$ is a countable collection of open sets in S . We claim that $\mathcal{B}$ is a basis for S . Let $U \subseteq S$ be open, and let $x \in U$. Then $\pi(U) \subseteq S/\sim$ is open, and hence there exists $B' \in \mathcal{B}'$ such that $[x] \in B' \subseteq \pi(U)$. Then we have $x \in \pi^{-1}(B') \subseteq U$, and hence $\mathcal{B}$ is a basis for S . Thus, S is second countable. ■

Example 1.40. Consider the disc $D^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1^2 + \dots + x_n^2 \leq 1\}$, and the hemisphere $H^n = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_{n+1} \geq 0\}$. On H^n , define an equivalence relation as $(x_1, \dots, x_n, x_{n+1}) \sim (y_1, \dots, y_n, y_{n+1})$ if either

1. $x_{n+1} = y_{n+1} = 0$ and $(y_1, \dots, y_n, 0) = (-x_1, \dots, -x_n, 0)$, or
2. $(x_1, \dots, x_n, x_{n+1}) = (y_1, \dots, y_n, y_{n+1})$.

On D^n , define an equivalence relation simply as $x \sim y$ if $x = y$ or $x = -y$. Then, one can show that $\mathbb{R}P^n \cong S^n / \sim \cong D^n / \sim \cong H^n / \sim$ as smooth manifolds, where $\sim$ on S^n is the antipodal equivalence relation.

1.5 Differential Structure

Let M be a smooth manifold of dimension n . For a point $p \in M$, let (U, φ) be a chart of M such that $p \in U$. The functions $r^1, \dots, r^n : \mathbb{R}^n \rightarrow \mathbb{R}$ defined as $r^i(x_1, \dots, x_n) = x_i$ are smooth, and hence, we can interpret (U, φ) as $(U, x^1, \dots, x^n)$, a chart with smooth coordinate functions $x^i = r^i \circ \varphi : U \rightarrow \mathbb{R}$ for $i = 1, \dots, n$. Then, given a map $f \in C^\infty(U)$, we define the partial derivative of f with respect to the coordinate function x^i at p as

$$\frac{\partial f}{\partial x^i}(p) := \frac{\partial(f \circ \varphi^{-1})}{\partial r^i}(\varphi(p)). \quad (1.66)$$

In fact, given any p in this particular U , we have

$$(\varphi^{-1})^* \left(\frac{\partial f}{\partial x^i} \right) = \frac{\partial f}{\partial x^i} \circ \varphi^{-1} = \frac{\partial(f \circ \varphi^{-1})}{\partial r^i} \in C^\infty(\varphi(U)). \quad (1.67)$$

This implies that $\frac{\partial f}{\partial x^i} \in C^\infty(U)$, and hence the partial derivative of f with respect to x^i is smooth on U . Moreover, one can show that $\frac{\partial x^i}{\partial x^j}(p) = \frac{\partial r^i}{\partial r^j}(\varphi(p)) = \delta_j^i$.

Let $F : N \rightarrow M$ be a smooth map between smooth manifolds of dimension n and m , respectively. Let (U, φ) be a chart on N with $p \in U$, and (V, ψ) be a chart on M with $F(p) \in V$. Then we can interpret (U, φ) as $(U, x^1, \dots, x^n)$, and (V, ψ) as $(V, y^1, \dots, y^m)$, where $x^i = r^i \circ \varphi : U \rightarrow \mathbb{R}$ and $y^j = r^j \circ \psi : V \rightarrow \mathbb{R}$ are smooth coordinate functions. Then, we have $F^i = y^i \circ F = r^i \circ \psi \circ F \in C^\infty(U)$ for $i = 1, \dots, m$. The *Jacobian matrix* of F at p , with respect to the charts (U, φ) and (V, ψ) , is defined as the $m \times n$ matrix

$$J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p) \right)_{1 \leq i \leq m, 1 \leq j \leq n}. \quad (1.68)$$

Each entry here can be written as

$$\frac{\partial F^i}{\partial x^j}(p) = \frac{\partial(\psi \circ F \circ \varphi^{-1})^i}{\partial r^j}(\varphi(p)) = \frac{\partial(r^i \psi F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial(y^i F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial(F^i \circ \varphi^{-1})}{\partial r^j}(\varphi(p)). \quad (1.69)$$

Recall the inverse function theorem, which states that if $F : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is smooth, and the Jacobian matrix $J_F(p)$ is invertible at $p \in U$, then there exists an open neighbourhood V of $F(p)$ such that $F : U \rightarrow V$ is a diffeomorphism. This motivates the following definition. Thus, if $m = n$ and $J_F(p)$ is invertible, then F is a local diffeomorphism at p ; $\psi \circ F \circ \varphi^{-1}$ is a local diffeomorphism at $\varphi(p)$ if and only if F is a local diffeomorphism at p .

Let M be a smooth manifold of dimension n , and let $p \in M$. Suppose $F^1, F^2, \dots, F^n$ are smooth functions on some open neighbourhood $U \subseteq M$ of p . Then consider $F = (F^1, \dots, F^n) : U \rightarrow \mathbb{R}^n$, which is a smooth map. Then $\left(\frac{\partial F^i}{\partial x^j}(p) \right)_{n \times n}$ is invertible if and only if F is a local diffeomorphism at p if and only if $F : W \rightarrow F(W) \subseteq \mathbb{R}^n$ is a diffeomorphism for some open neighbourhood $W \subseteq U$ of p . Then, on some open set around p , we can use $F^1, \dots, F^n$ as local coordinates. This motivates the following definition.

Definition 1.41. Let M be a smooth manifold, with $p \in M$. Recall $C_p^\infty(M) \equiv C_p^\infty$, the local ring of germs of smooth functions at p , which is also an $\mathbb{R}$ -algebra. Also define $T_p M$ to be the $\mathbb{R}$ -vector space of derivations at p ; that is, $T_p M = \{D : C_p^\infty \rightarrow \mathbb{R} \mid D \text{ is linear and satisfies the Leibniz rule}\}$. This $T_p M$ is called the *tangent space* of M at p .

August 12th.

We leave it as an exercise to check that $\frac{\partial}{\partial x^i}|_p$ is a derivation from C_p^∞ to $\mathbb{R}$, and hence $\frac{\partial}{\partial x^i}|_p \in T_p M$ for all valid i . Given a smooth map $F : N \rightarrow M$ between smooth manifolds, we can define the map $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ as $D \mapsto F_{*,p}(D)$, where $F_{*,p}(D)(f) = D(f \circ F)$ for all $f \in C_{F(p)}^\infty$. This is the pushforward of f at p . Moreover, if $G : M \rightarrow P$ is another smooth map, then $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p}$. The map $F_{*,p}$ is linear, and hence a linear transformation between the tangent spaces $T_p N$ and $T_{F(p)} M$.

If $F : N \rightarrow M$ were a diffeomorphism, then there exists a diffeomorphism $G : M \rightarrow N$ such that $G \circ F = \text{id}_N$ and $F \circ G = \text{id}_M$. Now look at the chain

$$T_p N \xrightarrow{F_{*,p}} T_{F(p)} M \xrightarrow{G_{*,F(p)}} T_{G(F(p))} N = T_p N. \quad (1.70)$$

Here, we have $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p} = (\text{id}_N)_{*,p} = \text{id}_{T_p N}$, and similarly $F_{*,p} \circ G_{*,F(p)} = \text{id}_{T_{F(p)} M}$. Thus, $F_{*,p}$ is a linear isomorphism between the tangent spaces $T_p N$ and $T_{F(p)} M$.

Let $p \in M$, a smooth manifold, and let (U, φ) be a chart containing p . Since $C_p^\infty(M) \equiv C_p^\infty(U)$, the diffeomorphism $\varphi : U \rightarrow \varphi(U)$ induces an isomorphism $\varphi_{*,p} : T_p M \rightarrow T_{\varphi(p)} \mathbb{R}^n$. We know that the tangent space on the right has the basis $\{\frac{\partial}{\partial x^1}|_{\varphi(p)}, \dots, \frac{\partial}{\partial x^n}|_{\varphi(p)}\}$. Applying this map to a basis element, we have

$$\varphi_{*,p} \left(\frac{\partial}{\partial x^i} \Big|_p \right) (f) = \frac{\partial}{\partial x^i} \Big|_p (f \circ \varphi) = \frac{\partial}{\partial r^i} \Big|_{\varphi(p)} (f \circ \varphi \circ \varphi^{-1}) = \frac{\partial}{\partial r^i} \Big|_{\varphi(p)} (f) \quad \text{for all } f \in C_p^\infty(M). \quad (1.71)$$

Hence, $\varphi_{*,p} \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial}{\partial r^i} \Big|_{\varphi(p)}$, and so $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ is a basis for $T_p M$. Thus, we have shown that the tangent space $T_p M$ has dimension n . This also shows that if $F : N \rightarrow M$ is a smooth map between smooth manifolds of dimension n and m that also happens to be a diffeomorphism, then $n = m$. Moreover, given a map $F : N \rightarrow M$ between smooth manifolds, and given a chart (U, φ) of $p \in N$ and a chart (V, ψ) of $F(p) \in M$, we have

$$F_{*,p} : T_p N \rightarrow T_{F(p)} M, \quad F_{*,p} \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \sum_{j=1}^m b_j^i \frac{\partial}{\partial y^j} \Big|_{F(p)}, \quad b_j^i = \frac{\partial F^j}{\partial x^i}(p) \quad \text{for } i = 1, \dots, n, j = 1, \dots, m. \quad (1.72)$$

That is, the matrix of $F_{*,p}$ with respect to the bases $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ and $\{\frac{\partial}{\partial y^1}|_{F(p)}, \dots, \frac{\partial}{\partial y^m}|_{F(p)}\}$ is precisely the $m \times n$ real *Jacobian matrix* $(dF)_p := J_F(p) = \left(\frac{\partial F^j}{\partial x^i}(p) \right)_{1 \leq i \leq n, 1 \leq j \leq m}$.

August 17th.

Suppose $0 \in (a, b) \subseteq \mathbb{R}$, and let $c : (a, b) \rightarrow M$ be a smooth map to a smooth manifold M of dimension n such that $c(0) = p \in M$. c is then called a *smooth curve* in M . Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M with $p \in U$. Then $c'(0) \in T_p M$ is defined as $c'(0) = c_{*,0} \left(\frac{d}{dt} \Big|_0 \right)$, where $\frac{d}{dt} \Big|_0 \in T_0 \mathbb{R}$ is the standard basis element. Then $c'(0)$ is a tangent vector at p , and we have

$$\mathbb{R} \cong T_0((a, b)) \xrightarrow{c_{*,0}} T_p M \xrightarrow{\varphi_{*,p}} T_{\varphi(p)} \mathbb{R}^n \cong \mathbb{R}^n, \quad c'(0) \mapsto \varphi_{*,p}(c'(0)) = (\varphi \circ c)_{*,0} \left(\frac{d}{dt} \Big|_0 \right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i} \Big|_p \quad (1.73)$$

since $\{\frac{\partial}{\partial x^i}|_p\}_{i=1}^n$ is a basis for $T_p M$. Utilising $c'(0) = c_{*,0} \left(\frac{d}{dt} \Big|_0 \right)$, we have

$$a^i = \varphi_{*,p}(c'(0))(x^i) = c_{*,0} \left(\frac{d}{dt} \Big|_0 \right) (x^i \circ \varphi^{-1}) = \frac{d}{dt} \Big|_0 (x^i \circ \varphi^{-1} \circ c) = \frac{d}{dt} \Big|_0 (x^i \circ c) = \frac{dc^i}{dt}(0). \quad (1.74)$$

Thus, a curve c can be thought of as a tangent vector at p , and the tangent vector is given by the derivative of the curve at 0. In particular, let us interpret $c : \mathbb{R} \rightarrow \mathbb{R}^n$ as a smooth curve in $\mathbb{R}^n$; we have $c(t) = (c_1(t), \dots, c_n(t))$, and hence $c'(0) = (\frac{dc_1}{dt}(0), \dots, \frac{dc_n}{dt}(0)) \in \mathbb{R}^n$. This is consistent with the above definition of $c'(0)$ as a tangent vector at p . From here, we wish to obtain a smooth map $c : (a, b) \rightarrow M$ such that $c(0) = p$ and $c'(0) = X_p = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i} \Big|_p$ for some $X_p \in T_p M$. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M with $p \in U$. In fact, choose a chart φ such that $\varphi(p) = 0 \in \mathbb{R}^n$ (translations are diffeomorphisms). Then we can define a smooth curve $c : (-\epsilon, \epsilon) \rightarrow M$ for some $\epsilon > 0$ such that $c(0) = p$ and $c'(0) = X_p$ as

$$c(t) = \varphi^{-1}(ta^1, \dots, ta^n) \quad \text{for } t \in (-\epsilon, \epsilon). \quad (1.75)$$

Then, $c'(0) = c_{*,0} \left(\frac{d}{dt} \Big|_0 \right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i} \Big|_p = X_p$, as desired.

Coming back to the smooth map $F : N \rightarrow M$ between smooth manifolds, the rank of F at $p \in N$ is defined as the rank of the linear map $F_{*,p} : T_p N \rightarrow T_{F(p)} M$, which is also the rank of the matrix $(dF)_p = J_F(p) = \left(\frac{\partial F^j}{\partial x^i}(p) \right)_{1 \leq i \leq n, 1 \leq j \leq m}$. The map $F : N \rightarrow M$ is called an *immersion* at $p \in N$ if $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ is injective (that is, $\text{rank}(F_{*,p}) = \dim(T_p N) = n$), and F is called an immersion if it is an immersion at all points $p \in N$. Similarly, F is called a *submersion* at $p \in N$ if $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ is surjective (that is, $\text{rank}(F_{*,p}) = \dim(T_{F(p)} M) = m$), and F is called a submersion if it is a submersion at all points $p \in N$. If $F : N \rightarrow M$ is a smooth map between smooth manifolds of dimension n and m , respectively, then $\text{rank}(F_{*,p}) \leq \min\{n, m\}$ for all $p \in N$. Thus, if $n < m$, then F cannot be a submersion, and if $n > m$, then F cannot be an immersion.

1.6 Regular Maps and Regular Submanifolds

Definition 1.42. Let $F : N \rightarrow M$ be a smooth map between smooth manifolds. $p \in N$ is called a *critical point* of F if $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ is not surjective; that is, F is not a submersion at p . p is called a *regular point* of F if $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ is surjective; that is, F is a submersion at p . Similarly, $q \in M$ is called a *critical value* of F if there exists a critical point $p \in N$ such that $F(p) = q$. $q \in M$ is called a *regular value* of F if it is not a critical value; that is, for all $p \in N$ such that $F(p) = q$, we have that p is a regular point of F . That is, either $F^{-1}(m) = \emptyset$, or $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ is surjective for all $p \in F^{-1}(m)$.

From here, we can define something called a regular submanifold, which is useful in determining whether a certain set is a manifold or not. Let M be a manifold of dimension n , and $S \subseteq M$ be a subset with the subspace topology. Suppose that for all $p \in S$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ of M with $p \in U$ and $\varphi(p) = 0 \in \varphi(U) \subseteq \mathbb{R}^n$ such that

$$U \cap S = \{(x^1, x^2, \dots, x^n) \in U \mid x^{i_1} = x^{i_2} = \dots = x^{i_{n-k}} = 0 \text{ for some } 1 \leq i_1 < i_2 < \dots < i_{n-k} \leq n\} \quad (1.76)$$

for some fixed integer $0 \leq k \leq n$ (via switching of coordinates, we may assume that the last $n - k$ coordinates are zero). Such a chart (U, φ) of M is said to be adapted to S . Then, we say that S is a *regular submanifold* of M of dimension k if for all $p \in S$, there exists a chart (U, φ) of M with $p \in U$ such that (U, φ) is adapted to S . Therefore, if (U, φ) is adapted to S , then $\varphi(U \cap S)$ consists of only those points that look like $(r^1, \dots, r^k, 0, \dots, 0) \in \mathbb{R}^n$, and hence $\varphi(U \cap S)$ is an open subset of $\mathbb{R}^k$. If we define $\varphi_S : U \cap S \rightarrow \mathbb{R}^k$ as $\varphi_S(p) = (x^1(p), \dots, x^k(p))$, then $\varphi_S(U \cap S) = \varphi(U \cap S) \cap \mathbb{R}^k$.

Theorem 1.43. Consider the maximal atlas $\mathcal{U} = \{(U, \varphi)\}$ of M , and let $\mathcal{A} = \{(U \cap S, \varphi_S) \mid (U, \varphi) \in \mathcal{U}\}$, where (U, φ) is adapted to S and $\bigcup \{U \cap S \mid (U, \varphi) \in \mathcal{U}\} = S$. Then $\mathcal{A}$ is a smooth atlas on S , and hence S is a smooth manifold of dimension k .

The *codimension* of a regular submanifold S of M is defined as $\text{codim}_M S = \dim(M) - \dim(S) = n - k$.

Let $f : M \rightarrow \mathbb{R}$ be a smooth function on a smooth manifold M of dimension n . Also let $0 \in \mathbb{R}$ be a regular value of f with $f^{-1}(0) \neq \emptyset$. Then, $f^{-1}(0)$ is a regular submanifold of M of dimension $n - 1$, or codimension 1. Note that 0 here is a red herring; we can replace 0 with any regular value $c \in \mathbb{R}$ of f with $f^{-1}(c) \neq \emptyset$, and the same result holds.

For a proof of this, let $p \in f^{-1}(0) \neq \emptyset$ so that $f(p) = 0$. Since 0 is a regular value of f , we have that $f_{*,p} : T_p M \rightarrow T_0 \mathbb{R}$ is surjective. $T_p M$ has a basis $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$, and $T_0 \mathbb{R}$ has a basis $\{\frac{d}{dt}|_0\}$. The $1 \times n$ Jacobian matrix of f at p is given by

$$(df)_p = \left(\frac{\partial f}{\partial x^1}(p), \dots, \frac{\partial f}{\partial x^n}(p) \right). \quad (1.77)$$

Since $f_{*,p} : T_p M \rightarrow T_0 \mathbb{R}$ is surjective, we have that $\text{rank}(f_{*,p}) = 1$, and hence $(df)_p \neq 0$; at least coordinate function x^i has $\frac{\partial f}{\partial x^i}(p) \neq 0$. Without the loss of generality, we may assume that $\frac{\partial f}{\partial x^1}(p) \neq 0$. Then, by the inverse function theorem, there exists an open neighbourhood V of p such that $f : V \rightarrow f(V) \subseteq \mathbb{R}$ is a diffeomorphism. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M with $p \in U \subseteq V$. Then, we have

$$U \cap f^{-1}(0) = \{(x^1, \dots, x^n) \in U \mid x^1 = 0\}, \quad (1.78)$$

which shows that (U, φ) is adapted to $f^{-1}(0)$. Since $p \in f^{-1}(0)$ was arbitrary, we have that $f^{-1}(0)$ is a regular submanifold of M of dimension $n - 1$, or codimension 1. In general, this result generalizes to the following theorem.

Theorem 1.44. Let $F : N \rightarrow M$ be a smooth map between smooth manifolds of dimension n and m , respectively, with $n \geq m$. Let $q \in M$ be a regular value of F with $F^{-1}(q) \neq \emptyset$. Then, $F^{-1}(q)$ is a regular submanifold of N of dimension $n - m$, or codimension m .

Proof. Let $p \in F^{-1}(q)$; then $F_{*,p} : T_p N \rightarrow T_q M$ is surjective. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart in N such that $\varphi(p) = 0 \in \mathbb{R}^n$, and let $(V, \psi) = (V, y^1, \dots, y^m)$ be a chart in M such that $\psi(q) = 0 \in \mathbb{R}^m$.

We may assume $F(U) \subseteq V$. The matrix of $F_{*,p}$ with respect to the bases $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ and $\{\frac{\partial}{\partial y^1}|_q, \dots, \frac{\partial}{\partial y^m}|_q\}$ is given by $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \leq i \leq m, 1 \leq j \leq n}$; this has rank m since $F_{*,p}$ is surjective. Without the loss of generality, we may assume that the $m \times m$ submatrix $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \leq i, j \leq m}$ is invertible. Then, by the inverse function theorem, there exists an open neighbourhood W of p such that $F : W \rightarrow F(W)$ is a diffeomorphism. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on N with $p \in U \subseteq W$. Then, we have

$$U \cap F^{-1}(q) = \{(x^1, \dots, x^n) \in U \mid x^1 = x^2 = \dots = x^m = 0\}, \quad (1.79)$$

which shows that (U, φ) is adapted to $F^{-1}(q)$. Since $p \in F^{-1}(q)$ was arbitrary, we have that $F^{-1}(q)$ is a regular submanifold of N of dimension $n - m$, or codimension m . ■

August 19th.

Example 1.45. Consider the map $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined as $f(x, y, z) = x^2 + y^2 + z^2$. Then, $S^2 = f^{-1}(1) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$. We have $f_{*,p} : T_p \mathbb{R}^3 \rightarrow T_1 \mathbb{R} = \mathbb{R}$ given by $f_{*,p}(\frac{\partial}{\partial x^i}|_p) = \sum_{j=1}^3 \frac{\partial f}{\partial x^j}(p) \frac{\partial}{\partial y^j}|_1$, where y is the coordinate function on $\mathbb{R}$. Then, the Jacobian matrix of f at p is given by $(df)_p = \left(\frac{\partial f}{\partial x}(p), \frac{\partial f}{\partial y}(p), \frac{\partial f}{\partial z}(p)\right) = (2x, 2y, 2z)$. The origin is the only critical point of f and is excluded from S^2 . Thus, $1 \in \mathbb{R}$ is a regular value of f , and hence $S^2 = f^{-1}(1)$ is a regular submanifold of $\mathbb{R}^3$ of dimension $3 - 1 = 2$.

Example 1.46. Consider the determinant map $\det : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$, interpreting a point $p \in \mathbb{R}^{n^2}$ as an $n \times n$ matrix $A \in M_n(\mathbb{R})$. Look at $SL_n(\mathbb{R}) = \det^{-1}(1) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}$. We have $\det_{*,A} : T_A \mathbb{R}^{n^2} \rightarrow T_1 \mathbb{R} = \mathbb{R}$, where the bases are $\{\frac{\partial}{\partial x_{ij}}\}_{1 \leq i, j \leq n}$ and $\{\frac{d}{dt}|_1\}$, respectively. Note that $\frac{\partial \det}{\partial x_{ij}}(A)$ is the (i, j) -th cofactor of A ; not all of these cofactors can be zero since otherwise we would have $\det(A) = 0$ but $A \in \det^{-1}(1)$. Thus, $1 \in \mathbb{R}$ is a regular value of $\det$, and hence $SL_n(\mathbb{R}) = \det^{-1}(1)$ is a regular submanifold of $\mathbb{R}^{n^2}$ of dimension $n^2 - 1$.

The following is a corollary of the inverse function theorem; it may be generalized to the case of smooth maps between smooth manifolds.

Corollary 1.47. Let $p \in \mathbb{R}^n$ have an open neighbourhood $U \subseteq \mathbb{R}^n$, and let $f : U \rightarrow \mathbb{R}^m$ be a smooth map with $f = (f_1, \dots, f_m)$. Suppose f has rank k in a neighbourhood of p in U ; that is, $\left(\frac{\partial f_i}{\partial x_j}(q)\right)$ has rank k for all q in this possibly smaller neighbourhood of p . Then there exists a diffeomorphism $G : V \rightarrow G(V)$, with $G(V)$ open in $\mathbb{R}^n$ and $p \in V$ open in U , and a diffeomorphism $F : W \rightarrow F(W)$, with $F(W)$ open in $\mathbb{R}^m$ and $f(p) \in W$ open in $\mathbb{R}^m$, such that $f(V) \subseteq W$, and $(FfG^{-1})(x^1, \dots, x^n) = (x^1, \dots, x^k, 0, \dots, 0)$ for all $(x^1, \dots, x^n) \in G(V)$.

Proof. Let $p \in \mathbb{R}^n$ have an open neighbourhood $U \subseteq \mathbb{R}^n$, and let $f : U \rightarrow \mathbb{R}^m$ be a smooth map with $f = (f_1, \dots, f_m)$. Suppose f has rank k in a neighbourhood of p in U . Then, without the loss of generality, we may assume that the $k \times k$ submatrix $\left(\frac{\partial f_i}{\partial x_j}(q)\right)_{1 \leq i, j \leq k}$ is invertible for all q in this possibly smaller neighbourhood of p . Define the smooth map $F : U \rightarrow \mathbb{R}^n$ as $F(x^1, \dots, x^n) = (f_1(x^1, \dots, x^n), \dots, f_k(x^1, \dots, x^n), x^{k+1}, \dots, x^n)$. Then, the Jacobian matrix of F at $q \in U$ is given by

$$(dF)_q = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(q) & \cdots & \frac{\partial f_1}{\partial x_k}(q) & \frac{\partial f_1}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_1}{\partial x_n}(q) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_k}{\partial x_1}(q) & \cdots & \frac{\partial f_k}{\partial x_k}(q) & \frac{\partial f_k}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_k}{\partial x_n}(q) \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.80)$$

Thus, by the inverse function theorem, there exists an open neighbourhood V of p with $V \subseteq U$ such that $F : V \rightarrow F(V)$ is a diffeomorphism. Consider the map $F^{-1} : F(V) \rightarrow V$; a point $(x^1, \dots, x^n) \in F(V)$ is

mapped to $(x^1, \dots, x^k, f^{k+1}, \dots, f^m)$. The Jacobian matrix of F^{-1} at $(x^1, \dots, x^n) \in F(V)$ is given by

$$(dF^{-1})_{(x^1, \dots, x^n)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ * & * & * & \frac{\partial f^{k+1}}{\partial x^{k+1}} & \dots & \frac{\partial f^{k+1}}{\partial x^n} \\ * & * & * & \vdots & \ddots & \vdots \\ * & * & * & \frac{\partial f^m}{\partial x^{k+1}} & \dots & \frac{\partial f^m}{\partial x^n} \end{pmatrix}. \quad (1.81)$$

We know this matrix to have rank k by hypothesis. Thus, the $(m-k) \times (n-k)$ submatrix $\left(\frac{\partial f^i}{\partial x^j}(q)\right)_{k+1 \leq i \leq m, k+1 \leq j \leq n}$ must have rank 0; that is, $\frac{\partial f^i}{\partial x^j}(q) = 0$ for all $k+1 \leq i \leq m$ and $k+1 \leq j \leq n$. The functions $f^{k+1}, \dots, f^m$ are then functions of only $x^1, \dots, x^k$. Define the smooth map $G : f(V) \rightarrow \mathbb{R}^m$ as $(y^1, \dots, y^m) = (y^1, \dots, y^k, y^{k+1} - f^{k+1}, \dots, y^m - f^m)$. Then, the Jacobian matrix of G at $(y^1, \dots, y^m) \in f(V)$ is given by

$$(dG)_{(y^1, \dots, y^m)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -\frac{\partial f^{k+1}}{\partial y^1} & \dots & -\frac{\partial f^{k+1}}{\partial y^k} & 1 & 0 & 0 \\ \vdots & \ddots & \vdots & 0 & 1 & 0 \\ -\frac{\partial f^m}{\partial y^1} & \dots & -\frac{\partial f^m}{\partial y^k} & 0 & 0 & 1 \end{pmatrix}. \quad (1.82)$$

This matrix is of full rank, so G is a diffeomorphism from $f(V)$ to $G(f(V))$. Then, we have $(GF)(x^1, \dots, x^n) = (f_1(x^1, \dots, x^n), \dots, f_k(x^1, \dots, x^n), 0, \dots, 0)$ for all $(x^1, \dots, x^n) \in V$. Thus, we have found diffeomorphisms G and F such that $(GF)(x^1, \dots, x^n) = (x^1, \dots, x^k, 0, \dots, 0)$ for all $(x^1, \dots, x^n) \in G(V)$. ■

The generalization is as follows. Let $f : N \rightarrow M$ be a smooth map between smooth manifolds of dimension n and m , respectively. Let $p \in N$ be a point such that f has a rank k in a neighbourhood of p in N . Then there exists a chart (U, φ) centred at p in N and a chart (V, ψ) centred at $f(p)$ in M such that $f(U) \subseteq V$, and $\psi \circ f \circ \varphi^{-1}(x^1, \dots, x^n) = (x^1, \dots, x^k, 0, \dots, 0)$ for all $(x^1, \dots, x^n) \in \varphi(U)$.

August 24th.

Theorem 1.48. *Let $f : N \rightarrow M$ be a smooth map between smooth manifolds of dimension n and m , respectively. Let $c \in M$ be such that f has a rank k in a neighbourhood of $f^{-1}(c) \neq \emptyset$ in N . Then $f^{-1}(c)$ is a regular submanifold of N of dimension $n - k$.*

Example 1.49. Look at the map $f : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ define by $f(A) = A^t A$. Then, $O_n(\mathbb{R}) = f^{-1}(I_n) = \{A \in GL_n(\mathbb{R}) \mid A^t A = I_n\}$. For now, the dimension of $O_n(\mathbb{R})$ will remain unknown, but we can still show that it is a regular submanifold of $GL_n(\mathbb{R})$. Let $C \in GL_n(\mathbb{R})$. Then $f(AC) = C^t f(A)C$. Note that conjugation by C is a diffeomorphism from $GL_n(\mathbb{R})$ to itself, and hence f has the same rank at A and AC . We shall show that the ranks of $f_{*,A}$ and $f_{*,B}$ are the same for all $A, B \in GL_n(\mathbb{R})$. We can break down $B = AC$ for some $C \in GL_n(\mathbb{R})$. Then $f \circ r_C = l_{C^t} \circ r_C \circ f$, where $r_C : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is right multiplication by C and $l_{C^t} : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is left multiplication by C^t . Then, we have

$$f_{*,AC}(r_C)_{*,A} = (l_{C^t})_{*,A^t AC}(r_C)_{*,A^t A} f_{*,A}. \quad (1.83)$$

The rank of $f_{*,A}$ is the same as the rank of $f_{*,AC}$ since $(r_C)_{*,A}$ and $(l_{C^t})_{*,A^t AC}(r_C)_{*,A^t A}$ are both isomorphisms. Thus, f has the same rank at all points in $GL_n(\mathbb{R})$.

The previous theorem may be generalized to the following theorem.

Theorem 1.50. *Let $f : N \rightarrow M$ be a smooth map between smooth manifolds of dimension n and m , respectively. Also let $S \subseteq M$ be a regular submanifold of codimension k , or dimension $m - k$. Suppose $f_{*,p}(T_p N) + T_{f(p)} S = T_{f(p)} M$ for all $p \in f^{-1}(S)$. Then, $f^{-1}(S)$ is a regular submanifold of N of*

codimension k , or dimension $n - k$.

Theorem 1.51. *Let $f : N \rightarrow M$ be a smooth map between smooth manifolds of dimension n and m , respectively. Let $p \in N$.*

1. *Suppose $\text{rank}(f_{*,p}) = m$. Then there exists a local chart (U, φ) centred at p in N and a local chart (V, ψ) centred at $f(p)$ in M such that $\psi f \varphi^{-1}(x^1, \dots, x^n) = (x^1, \dots, x^m)$ for all $(x^1, \dots, x^n) \in \varphi(U)$.*
2. *Suppose $\text{rank}(f_{*,p}) = n$. Then there exists a local chart (U, φ) centred at p in N and a local chart (V, ψ) centred at $f(p)$ in M such that $\psi f \varphi^{-1}(x^1, \dots, x^n) = (x^1, \dots, x^n, 0, \dots, 0)$ for all $(x^1, \dots, x^n) \in \varphi(U)$.*

Chapter 2

TANGENT SPACES, BUNDLES, AND VECTORS

2.1 Manifold Structure

August 31st.

Recall that given a smooth manifold M , the tangent space $T_p M$ at a point $p \in M$ consists of all derivations from $C_p^\infty(M)$ to $\mathbb{R}$. $T_p M$ exactly equals $T_p U$, where U is an open neighbourhood of p in M , since germs of functions need only match in a neighbourhood. If M is a manifold of dimension n , then any tangent space of a point in M is a vector space of dimension n , too. Now, the *tangent bundle* of the manifold is defined as the disjoint union of all tangent spaces:

$$TM = \bigsqcup_{p \in M} T_p M. \quad (2.1)$$

From here, let $\pi : TM \rightarrow M$ denote the projection map; that is, if $v \in T_p M \subseteq TM$ for some $p \in M$, then $\pi(v) = p$.

We can equip TM with a topology so that it becomes a smooth manifold in its own right, and so that π also happens to be a smooth map. For $p \in M$, let (U, φ) be a chart on M containing p . Note that for all $p \in U$, the tangent space $T_p M$ has a basis $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$. Given any $v \in T_p M$, it can uniquely be written as

$$v = \sum_{i=1}^n c^i(v) \frac{\partial}{\partial x^i} \Big|_p. \quad (2.2)$$

Now, given $v \in TU$, we define a chart $(\tilde{U}, \tilde{\varphi})$ as $\tilde{U} = \pi^{-1}(U) = \bigsqcup_{p \in U} T_p M = TU$, and

$$\tilde{\varphi} : TU \rightarrow \varphi(U) \times \mathbb{R}^n, \quad v \mapsto (x^1(p), \dots, x^n(p), c^1(v), \dots, c^n(v)) = ((x^1 \circ \pi)(v), \dots, (x^n \circ \pi)(v), c^1(v), \dots, c^n(v)). \quad (2.3)$$

Note that $\tilde{\varphi}$ has a (set-theoretic) inverse since

$$\tilde{\varphi}^{-1} : \varphi(U) \times \mathbb{R}^n \rightarrow TU, \quad (\varphi(p), c^1, \dots, c^n) \mapsto v = \sum_{i=1}^n c^i \frac{\partial}{\partial x^i} \Big|_p. \quad (2.4)$$

On $\varphi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$, we put the subspace topology. Using the bijection $\tilde{\varphi}$, we put a topology on TU ; that is, $A \subseteq TU$ is open if and only if $\tilde{\varphi}(A) \subseteq \varphi(U) \times \mathbb{R}^n$ is open.

With $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}$ is a maximal atlas of M , let $\mathcal{B} = \{A \subseteq TU_\alpha \mid A \text{ open and } \alpha \text{ is any index}\}$. We show that $\mathcal{B}$ is a basis for a topology on TM . The union being M is clear since

$$TM = \bigcup_{\alpha} TU_\alpha \subseteq \bigcup_{A \in \mathcal{B}} A \subseteq TM. \quad (2.5)$$

To show the second property of a basis, let $A \subseteq TU$ be open and $B \subseteq TV$ be open. Then $TU \cap TV = \pi^{-1}(U) \cap \pi^{-1}(V) = \pi^{-1}(U \cap V) = T(U \cap V)$ and so $A \cap B \subseteq T(U \cap V)$. Since $A \cap T(U \cap V)$ and $B \cap T(U \cap V)$ are both open in $T(U \cap V)$, $A \cap B = A \cap B \cap T(U \cap V)$ is open in $T(U \cap V)$.

Lemma 2.1. *TM is a second countable Hausdorff space.*

Proof. Let $v, w \in TM$; so that $v \in T_p M$ and $w \in T_q M$, or $\pi(v) = p$ and $\pi(w) = q$. Two cases arise. If $p \neq q$, then since M is Hausdorff, there exists open neighbourhoods $p \in U$ and $q \in V$ such that $U \cap V = \emptyset$. This simply implies $v \in \pi^{-1}(U) = TU$ and $w \in \pi^{-1}(V) = TV$, and $TU \cap TV = \emptyset$. For the case when $p = q$, both $v, w \in T_p M$. Since this $T_p M$ identifies with $\mathbb{R}^n$, n being the dimension of M , we can pick open neighbourhoods $U_1, U_2 \subseteq T_p M$ of v and w , respectively, such that $U_1 \cap U_2 = \emptyset$. If we now pick a coordinate chart W of $p \in M$, then $W \times U_1$ and $W \times U_2$ are two open sets in TM that separate v and w , with $W \times U_1 \cap W \times U_2 = \emptyset$.

Let $\{B_i\}_{i \geq 1}$ be a countable basis of M , and let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ be an atlas. For $p \in U_\alpha$, there exists $p \in B_{p,\alpha} \subseteq U_\alpha$. Varying p and α , we get $\{B_{p,\alpha}\}_{p \in M, \alpha \in \Lambda} \subseteq \{B_i\}_{i \geq 1}$. So, there exists a countable basis of M consisting of coordinate charts for M . Since TU_i identifies with $\varphi_i(U_i) \times \mathbb{R}^n$ for any i , it is second countable and has a countable basis $\mathcal{B}_i$. Therefore, $\mathcal{B} = \bigcup_{i=1}^\infty \mathcal{B}_i$ is a countable basis of TM . ■

Finally, we show that $\{(TU_\alpha, \tilde{\varphi}_\alpha)\}$ gives an atlas consisting of smooth compatible charts on TM . Let $(U, \varphi) = (U, x^1, \dots, x^n)$ and $(V, \psi) = (V, y^1, \dots, y^n)$ be two charts in M such that $U \cap V \neq \emptyset$. Let $p \in U \cap V$ with $v \in T_p M$. $\{\frac{\partial}{\partial x^i}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ and $\{\frac{\partial}{\partial y^1}|_p, \dots, \frac{\partial}{\partial y^n}|_p\}$ are two bases for $T_p M$, so $v = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}|_p = \sum_{i=1}^n b^i \frac{\partial}{\partial y^i}|_p$. Applying y^i on both sides gives

$$b^i = \sum_{j=1}^n \frac{\partial y^i}{\partial x^j}(p) a^j. \quad (2.6)$$

This is a smooth transformation, showing that switching from one set of coordinates to another is smooth. Moreover, the TU_α 's cover TM . This is a valid atlas.

As a consequence of this, the projection map $\pi : TM \rightarrow M$ is a smooth map, since $\varphi(U) \times \mathbb{R}^n \cong TU \xrightarrow{\pi} U \cong \varphi(U)$ takes the first n coordinates of $\varphi(U) \times \mathbb{R}^n$, which is a smooth map.

Definition 2.2. A *vector field* X on a smooth manifold M is a map $X : M \rightarrow TM$ such that $X(p) \in T_p M$ for all $p \in M$. In other words, a vector field is a smooth assignment of a tangent vector to each point in the manifold. It is termed smooth if the map X is smooth, meaning that for any chart (U, φ) on M , the composition $\tilde{\varphi} \circ X|_U : U \rightarrow \varphi(U) \times \mathbb{R}^n$ is smooth.

Such a vector field X is a section of the tangent bundle TM , as it assigns to each point $p \in M$ a vector in the corresponding tangent space $T_p M$. The projection map $\pi : TM \rightarrow M$ turns out to be the left inverse of X , since $\pi(X(p)) = p$ for all $p \in M$.

Definition 2.3. Let M be a smooth manifold of dimension n . A *smooth vector bundle* on M of rank r is a triple (E, M, π) , where E is a smooth manifold of dimension $n + r$, and $\pi : E \rightarrow M$ is a smooth surjective map such that for each $p \in M$, the fiber $\pi^{-1}(p)$ is an r -dimensional real vector space. Additionally, π satisfies the local triviality condition: for every $p \in M$, there exists an open neighborhood U of p in M and a diffeomorphism $\phi_U : E_U := \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$ such that

$$\pi|_{E_U} = \text{pr}_1 \circ \phi_U, \quad (2.7)$$

where $\text{pr}_1 : U \times \mathbb{R}^r \rightarrow U$ is the projection onto the first factor, and $\varphi_U|_{\pi^{-1}(p)} : \pi^{-1}(p) \rightarrow \{p\} \times \mathbb{R}^r$ is a linear isomorphism of real vector spaces for each $p \in U$. Here, E is termed the *total space* and M the *base space* of the vector bundle.

Definition 2.4. A *morphism* between two smooth vector bundles (E, M, π) and (F, N, φ) is a pair of smooth maps $(\tilde{f}, f)$, where $\tilde{f} : E \rightarrow F$ and $f : M \rightarrow N$, such that $\varphi \tilde{f} = f \pi$, and for each $p \in M$, the map $\tilde{f}|_{\pi^{-1}(p)} : \pi^{-1}(p) \rightarrow \varphi^{-1}(f(p))$ is a linear map of real vector spaces.

As a special case, if $M = N$, and $f = \text{id}_M$, then $\tilde{f}$ is called a *bundle map* over M . If $\tilde{f}$ is a diffeomorphism, then $(\tilde{f}, \text{id}_M)$ is an isomorphism of vector bundles, and the two bundles are said to be isomorphic. In particular, if (E, M, π) is isomorphic to $(M \times \mathbb{R}^r, M, \text{pr}_1)$, then (E, M, π) is called a *trivial vector bundle* of rank r .

2.1.1 Sections and Frames

September 2nd.

As earlier with a vector field, we have the following.

Definition 2.5. A map $s : M \rightarrow E$ is called a *section* of the vector bundle (E, M, π) if $\pi(s(p)) = p$ for all $p \in M$. In other words, a section assigns to each point in the base space a vector in the corresponding fiber. A section is termed smooth if the map s is smooth, meaning that for any chart (U, φ) on M , the composition $\phi_U \circ s|_U : U \rightarrow U \times \mathbb{R}^r$ is smooth.

We use $\Gamma(E) = \Gamma(M, E)$ to denote the set of all (global) smooth sections of the vector bundle (E, M, π) . For any open subset $U \subseteq M$, we denote by $\Gamma(U, E)$ the set of all smooth sections of (E, M, π) over U ; that is, smooth maps $s : U \rightarrow E$ such that $\pi(s(p)) = p$ for all $p \in U$. Moreover, for $s, t \in \Gamma(U, E)$, we define the sum $s + t \in \Gamma(U, E)$ by $(s + t)(p) = s(p) + t(p)$ for all $p \in U$; this sum is indeed in $\Gamma(U, E)$ since $\pi((s + t)(p)) = \pi(s(p) + t(p)) = \pi(s(p)) = p$, and $s + t$ can also be shown to be smooth. For $f \in C^\infty(U)$, we define the product $fs \in \Gamma(U, E)$ by $(fs)(p) = f(p)s(p)$ for all $p \in U$; this product is also in $\Gamma(U, E)$ since $\pi((fs)(p)) = \pi(f(p)s(p)) = \pi(s(p)) = p$, and fs can also be shown to be smooth. Therefore, $\Gamma(U, E)$ is a module over the ring $C^\infty(U)$, and $\Gamma(E)$ is a module over the ring $C^\infty(M)$.

Definition 2.6. A *frame* of a vector bundle (E, M, π) over an open subset $U \subseteq M$ is a collection of r sections $s_1, \dots, s_r \in \Gamma(U, E)$ such that for each $p \in U$, the set $\{s_1(p), \dots, s_r(p)\}$ forms a basis of the fiber $E_p = \pi^{-1}(p)$. The frame is termed smooth if each section s_i is smooth.

Consider the tangent bundle (TM, M, π) of a smooth manifold M of dimension n . Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M . Then the collection of sections $s_i : U \rightarrow TM$ defined by $s_i(p) = \frac{\partial}{\partial x^i} \Big|_p$ for $i = 1, \dots, n$ forms a smooth frame of the tangent bundle over U .

Lemma 2.7. Let (E, M, π) be a vector bundle, and let $U \subseteq M$ be an open subset. Suppose $s_1, \dots, s_r$ is a smooth frame of E over U , and $s = \sum_{i=1}^r c^i s_i$. Then s is a smooth section of E over U if and only if each coefficient function $c^i : U \rightarrow \mathbb{R}$ is smooth.

Proof. The “if” direction is clear from the definition of smoothness. For the “only if” direction, let s be a smooth section of E over U such that $s = \sum_{i=1}^r c^i s_i$. Let $p \in U$. We show that each c^i is smooth at p . Since $s_1, \dots, s_r$ is a smooth frame of E over U , there exists an open neighborhood $V \subseteq U$ of p and a diffeomorphism $\phi_V : E_V \rightarrow V \times \mathbb{R}^r$ such that $\pi|_{E_V} = \text{pr}_1 \circ \phi_V$, and $\phi_V|_{\pi^{-1}(q)} : \pi^{-1}(q) \rightarrow \{q\} \times \mathbb{R}^r$ is a linear isomorphism for each $q \in V$. Let $\phi_V(s(q)) = (q, (c^1(q), \dots, c^r(q)))$ for all $q \in V$. Since $\phi_V \circ s|_V : V \rightarrow V \times \mathbb{R}^r$ is smooth, it follows that each component function $c^i : V \rightarrow \mathbb{R}$ is smooth. Therefore, each coefficient function c^i is smooth at p , and since p was arbitrary, each c^i is smooth on all of U . ■

September 4th.

Definition 2.8. Let $f : N \rightarrow M$ be a smooth map between smooth manifolds, such that f is also injective and an immersion. Then $f(N)$, equipped with the topology borrowed from N , is termed an *immersed submanifold*.

In such a case, $f(N)$ is diffeomorphic to N . Note, however, that $f(N)$ need not be a manifold in the subspace topology of M ; the subspace topology on $f(N)$ is weaker than the topology on $f(N)$ borrowed from N .

Definition 2.9. A smooth map $f : N \rightarrow M$ between smooth manifolds is called an *embedding* if f is injective and an immersion, and the subspace topology from M and the topology inherited from N are the same on $f(N)$.

Theorem 2.10. If $S \subseteq M$ is a regular submanifold of M , then the inclusion map $i : S \rightarrow M$ is an embedding. Conversely, if $f : N \rightarrow M$ is an embedding, then $f(N)$ is a regular submanifold of M .

Proof. Note that $i : S \rightarrow M$ is a smooth map, taking $(x^1, x^2, \dots, x^s) \mapsto (x^1, x^2, \dots, x^s, 0, \dots, 0)$, is injective and an immersion. Also, the topologies trivially coincide. Hence, i is an embedding. For the converse, suppose $f : N \rightarrow M$ is an embedding. Let $p \in N$, with $f(p) \in f(N)$. Since f is an immersion (at p), there exist charts $(U, \varphi) = (U, x^1, \dots, x^n)$, centred at p , and $(V, \psi) = (V, y^1, \dots, y^m)$, centred at $f(p)$, such that $f(U) \subseteq V$, and f acts as $(x^1, \dots, x^n) \mapsto (x^1, \dots, x^n, 0, \dots, 0)$. Now, since $f(U) \subseteq f(N) \cap V$, we will change V to be the smallest open set $W \subseteq V \subseteq M$ such that $f(U) = f(N) \cap W$; since U is open in N if and only if $f(U)$ is open in $f(N)$ if and only if $f(U) = V' \cap f(N)$ where V' is open in M , we can set $W = V \cap V' \subseteq V$, so that $f(U) = W \cap f(N)$. ■

Theorem 2.11. Let $f : N^n \rightarrow M^m$ be a smooth map between smooth manifolds. Suppose $S^s \subseteq M$ is a regular submanifold of M , and $f(N) \subseteq S$; this induces a set-theoretic factoring map $\tilde{f} : N \rightarrow S$ such that $i \circ \tilde{f} = f$. Then $\tilde{f}$ is a smooth map between smooth manifolds.

Proof. With $p \in N$, and $f(p) \in S \subseteq M$, there exists a chart $(V, \psi) = (V, y^1, \dots, y^m)$, centred at $f(p)$ in M , such that $S \cap V = \{(y^1, \dots, y^m) \in V \mid y^{s+1} = \dots = y^m = 0\}$. There also exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$, centred at $p \in N$, such that $f(U) \subseteq V$ and f is given by $(x^1, \dots, x^n) \mapsto (f^1, f^2, \dots, f^m)$, where each f^i is smooth on U . But this means $f^{s+1}(x^1, \dots, x^n) = \dots = f^m(x^1, \dots, x^n) = 0$ for all $(x^1, \dots, x^n) \in U$. In S , we have a smooth chart given by $(V \cap S, \varphi_S)$ via which, the map $\tilde{f}$ on U is given as $(x^1, \dots, x^n) \mapsto (f^1, \dots, f^s)$; $\tilde{f}$ is smooth. ■

2.2 Bump Functions and Partitions of Unity

September 18th.

Definition 2.12. A *bump function* is a smooth non-negative function f on a smooth manifold M such that f has compact support, that is, the closure of the set $\{q \in M \mid f(q) \neq 0\}$ is compact in M . A bump function is said to be adapted to an open set $U \subseteq M$ if the support of f is contained in U .

If $p \in U \subseteq M$ is an open set, then there exists a bump function f adapted to U such that $f(x) = 1$ in a neighbourhood of p contained in U .

Definition 2.13. A collection of subsets $\{A_\alpha\}_{\alpha \in \Lambda}$ of a topological space S is said to be *locally finite* if for all $p \in S$, there exists a neighbourhood V of P such that $\{\alpha \in \Lambda \mid A_\alpha \cap V \neq \emptyset\}$ is finite. In other words, each point in S has a neighbourhood that intersects only finitely many of the sets A_α .

Definition 2.14. A (smooth) *partition of unity* is a collection of smooth functions $\{\rho_\alpha : M \rightarrow \mathbb{R}\}_{\alpha \in \Lambda}$ on a smooth manifold M such that the following conditions hold:

1. $\{\text{supp } \rho_\alpha\}_{\alpha \in \Lambda}$ is a locally finite collection of subsets of M .
2. $\sum_{\alpha \in \Lambda} \rho_\alpha(p) = 1$ for all $p \in M$.

Note that the sum in the second condition is well-defined since for each $p \in M$, there exists a neighbourhood V of p such that $\{\alpha \in \Lambda \mid \text{supp } \rho_\alpha \cap V \neq \emptyset\}$ is finite, and so only finitely many terms in the sum are non-zero.

Theorem 2.15. *Let M be a smooth manifold, and let $\{U_\alpha\}_{\alpha \in \Lambda}$ be an open cover of M . Then the following hold true.*

1. *There exists a countable smooth partition of unity $\{\varphi_i\}_{i=1}^\infty$ on M such that for each i , $\text{supp}(\varphi_i)$ is compact and contained in some U_α .*
2. *There exists a smooth partition of unity $\{\rho_\alpha\}_{\alpha \in \Lambda}$ on M such that for each α , $\text{supp}(\rho_\alpha)$ is contained in U_α .*

In the second statement, we say that the partition of unity is *subordinate* to the open cover $\{U_\alpha\}_{\alpha \in \Lambda}$. Before we prove the theorem, we require a few results from topology.

Firstly, since M is a smooth manifold, it is second countable and has a countable basis $\{B_i\}_{i \geq 1}$. Define $\mathcal{S} = \{B_i \mid \overline{B_i} \text{ is compact}\}$. Then $\mathcal{S}$ is a countable basis of M consisting of open sets with compact closure. We shall show that $\mathcal{S}$ also happens to be a basis of M ; to this end, let $p \in U \subseteq M$ be a point in an open set U . Moreover, let V be an open neighbourhood of p such that $\overline{V}$ is compact and $\overline{V} \subseteq U$. Since $\mathcal{B}$ is a basis of M , there exists $B_i \in \mathcal{B}$ such that $p \in B_i \subseteq V$. But then $\overline{B_i} \subseteq \overline{V}$ is compact, and so $B_i \in \mathcal{S}$. Therefore, $\mathcal{S}$ is a basis of M consisting of open sets with compact closure. Thus, any smooth manifold has a countable basis of open sets with compact closure.

Secondly, given any smooth manifold M , there exists a sequence of open sets $\{V_i\}_{i \geq 1}$ such that $\overline{V_i}$ is compact and $\overline{V_i} \subseteq V_{i+1}$ for all i , and $M = \bigcup_{i=1}^\infty V_i$. Indeed, let $\{B_i\}_{i=1}^\infty$ be a countable basis of M consisting of open sets with compact closure. Define $V_1 = B_1$, so that $V_1 \subseteq \overline{V_1}$ is compact. Thus, it must be contained in a finite union of the open cover $\{U_\alpha\}_{\alpha \in \Lambda}$; that is, there exists a finite set $A_1 \subseteq \Lambda$ such that $\overline{V_1} \subseteq \bigcup_{\alpha \in A_1} U_\alpha$; for simplicity, suppose $\overline{V_1} \subseteq B_1 \cup B_2 \cup \dots \cup B_{i_1}$ for some $i_1 > 1$ such that i_1 is the smallest integer with this property. Now, let $V_2 = B_1 \cup B_2 \cup \dots \cup B_{i_1}$, an open set with compact closure. Thus, $V_1 \subseteq \overline{V_1} \subseteq V_2$, and $\overline{V_2}$ is compact. Repeating the same argument, we have $\overline{V_2} \subseteq B_1 \cup B_2 \cup \dots \cup B_{i_1} \cup \dots \cup B_{i_2}$ for some $i_2 > i_1$ such that i_2 is the smallest integer with this property. Let $V_3 = B_1 \cup B_2 \cup \dots \cup B_{i_2}$, an open set with compact closure. Thus, $V_2 \subseteq \overline{V_2} \subseteq V_3$, and $\overline{V_3}$ is compact. Continuing in this manner, we get a sequence of open sets $\{V_i\}_{i=1}^\infty$ such that $\overline{V_i}$ is compact and $\overline{V_i} \subseteq V_{i+1}$ for all i , and $M = \bigcup_{i=1}^\infty V_i$. If we define $V_{-1} = V_0 = \emptyset$, we obtain a sequence as

$$V_{-1} \subseteq \overline{V_{-1}} \subseteq V_0 \subseteq \overline{V_0} \subseteq V_1 \subseteq \overline{V_1} \subseteq V_2 \subseteq \overline{V_2} \subseteq V_3 \subseteq \dots \quad (2.8)$$

From here, we can infer that

$$\overline{V_{i+1}} \setminus V_i \subseteq V_{i+2} \setminus \overline{V_{i-1}} \quad (2.9)$$

where $\overline{V_{i+1}} \setminus V_i$ is compact and $\overline{V_{i-1}} \subseteq V_{i+2}$ is open. We shall use this construction in the proof of the theorem.

Proof. Start with the construction of the sequence of open sets $\{V_i\}_{i=-1}^\infty$ as above, where $V_{-1} = V_0 = \emptyset$, and $\overline{V_i} \subseteq V_{i+1}$ for all i . Let $p \in U_\alpha$ for some $\alpha \in \Lambda$. Then there exists i such that $p \in U_\alpha \cap (V_{i+2} - \overline{V_{i-1}})$. Since $U_\alpha \cap (V_{i+2} - \overline{V_{i-1}})$ is open, there exists a smooth bump function $\rho_p : M \rightarrow \mathbb{R}$ such that $\text{supp}(\rho_p) \subseteq U_\alpha \cap (V_{i+2} - \overline{V_{i-1}})$ and $\rho_p > 0$ in some neighbourhood W_p of p . The collection $\{W_p\}_{p \in \overline{V_{i+1}} \setminus V_i}$ is then an open cover of the compact set $\overline{V_{i+1}} \setminus V_i$; refine this cover to a finite subcover consisting of $W_1, \dots, W_m$, and let $\rho_1, \dots, \rho_m$ be the corresponding bump functions. Repeating this for every i , we obtain, for each $\overline{V_{i+1}} \setminus V_i$, a finite cover $W_1^i, W_2^i, \dots, W_{m(i)}^i$ of $\overline{V_{i+1}} \setminus V_i$, and corresponding bump functions $\rho_1^i, \rho_2^i, \dots, \rho_{m(i)}^i$ such that $\text{supp}(\rho_j^i) \subseteq U_\alpha \cap (V_{i+2} - \overline{V_{i-1}})$ for some $\alpha \in \Lambda$, and $\rho_j^i \equiv 1$ in W_j^i for all $j = 1, \dots, m(i)$. Now, let $\{\rho_k\}_{k=1}^\infty$ be the countable collection of all these bump functions, and $\{W_j^i\}$ be the corresponding open sets. Note that the following properties hold:

1. $\text{supp}(\rho_j^i)$ is compact for all i, j .
2. $\text{supp}(\rho_j^i) \subseteq U_{\alpha_{ij}} \cap (V_{i+2} - \overline{V_{i-1}})$ for some $\alpha_{ij} \in \Lambda$.
3. $\overline{V_{i+1}} \setminus V_i \subseteq \bigcup_{j=1}^{m(i)} W_j^i$ for all i .
4. $\rho_j^i > 0$ on W_j^i for all i, j .

Since $\text{supp}(\rho_j^i) \subseteq V_{i+2} \setminus \overline{V_{i-1}}$, we must have $\text{supp}(\rho_j^i) \cap \overline{V_{i-1}} = \emptyset$ and $\text{supp}(\rho_j^i) \cap (M \setminus V_{i+2}) = \emptyset$. Therefore, for each $p \in M$, there exists i such that $p \in V_{i+2} \setminus \overline{V_{i-1}}$, and so p can only belong to the support of bump

functions ρ_j^k for $k = i - 1, i, i + 1$. This shows that the collection $\{\text{supp}(\rho_j^i)\}$ is locally finite. Finally, we define

$$\varphi_k = \frac{\rho_k}{\sum_{j=1}^{\infty} \rho_j} \quad (2.10)$$

where the denominator is smooth and positive, and only finitely many terms are non-zero in a neighbourhood of any point. Then $\{\varphi_k\}_{k=1}^{\infty}$ is a countable smooth partition of unity on M such that for each k , $\text{supp}(\varphi_k)$ is compact and contained in some U_{α} . This proves the first statement. For the second statement, we can define $\rho_{\alpha} = \sum_{k \in I_{\alpha}} \varphi_k$, where $I_{\alpha} = \{k \mid \text{supp}(\varphi_k) \subseteq U_{\alpha}\}$. Then $\{\rho_{\alpha}\}_{\alpha \in \Lambda}$ is a smooth partition of unity on M subordinate to the open cover $\{U_{\alpha}\}_{\alpha \in \Lambda}$. ■

September 21st.

Lemma 2.16. *Let M be a smooth manifold, and X be a vector field on M . Then X is smooth on M if and only if for any chart (U, φ) on M , with $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$ on U , each coefficient function $a^i : U \rightarrow \mathbb{R}$ is smooth.*

Proof. Suppose X was smooth on M , or $X : M \rightarrow TM$ is a smooth map. Then for any chart (U, φ) on M , the composition $\tilde{\varphi} \circ X|_U : U \rightarrow \varphi(U) \times \mathbb{R}^n$ is smooth, and so each component function $a^i : U \rightarrow \mathbb{R}$ is smooth. Conversely, suppose that for any chart (U, φ) on M , with $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$ on U , each coefficient function $a^i : U \rightarrow \mathbb{R}$ is smooth. Then for any chart (U, φ) on M , the composition $\tilde{\varphi} \circ X|_U : U \rightarrow \varphi(U) \times \mathbb{R}^n$ is smooth, and so $X : M \rightarrow TM$ is a smooth map. ■

Proposition 2.17. *Let X be a vector field on a smooth manifold M . Then X is smooth if and only if for any smooth function $f \in C^{\infty}(M)$, the function $X(f) : M \rightarrow \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth.*

Proof. Suppose X is smooth. Then for any smooth function $f \in C^{\infty}(M)$, the function $X(f) : M \rightarrow \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. Conversely, suppose that for any smooth function $f \in C^{\infty}(M)$, the function $X(f) : M \rightarrow \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. Let (U, φ) be a chart on M , with $\varphi = (x^1, \dots, x^n)$. Then on U , we have $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$, where each coefficient function $a^i : U \rightarrow \mathbb{R}$ is given by $a^i(p) = X_p(x^i)$. Since each coordinate function x^i is smooth, it follows that each coefficient function $a^i : U \rightarrow \mathbb{R}$ is smooth. Therefore, by the previous lemma, X is smooth on M . ■

Moreover, since X_p is a $\mathbb{R}$ -linear derivation from $C^{\infty}(M)$ to $\mathbb{R}$, we get that X is a derivation on $C^{\infty}(M)$; that is, for any $f, g \in C^{\infty}(M)$, we have

$$X(fg)(p) = X_p(fg) = f(p)X_p(g) + g(p)X_p(f) = (fX(g) + gX(f))(p) \implies X(fg) = fX(g) + gX(f). \quad (2.11)$$

2.3 Integral Curves

The following theorem will be useful in the study of vector fields and their flows on smooth manifolds.

Theorem 2.18. *Let $p_0 \in V \subseteq \mathbb{R}^n$ be an open set, and $f : V \rightarrow \mathbb{R}^n$ be a smooth map. Then the differential equation $\frac{dy}{dt} = f(y)$ with initial condition $y(0) = p_0$ has a unique smooth solution $y : (a(p_0), b(p_0)) \rightarrow V$ where $(a(p_0), b(p_0))$ is the maximal open interval containing 0 such that y is defined on $(a(p_0), b(p_0))$.*

Definition 2.19. Let X be a smooth vector field on a smooth manifold M , and let $p \in M$. An *integral curve* of X is a smooth curve $c : (a, b) \rightarrow M$ such that $c'(t) = X_{c(t)}$ for all $t \in (a, b)$. We usually presume $0 \in (a, b)$; if $c(0) = p$, we say that c is an integral curve of X starting at p . A *maximal integral curve* is an integral curve whose domain cannot be extended to a larger interval while still satisfying the integral curve condition.

We look at a few examples.

Example 2.20. Consider the smooth manifold $M = \mathbb{R}^2$ and the smooth vector field $X = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$. Let $p_0 = (1, 0) \in \mathbb{R}^2$. Then, an integral curve $c : (a, b) \rightarrow \mathbb{R}^2$ of X starting at p_0 satisfies the differential equation $c'(t) = (\dot{x}(t), \dot{y}(t)) = (-y(t), x(t)) = X_{c(t)}$ with initial condition $c(0) = (1, 0)$. Solving this system of differential equations, we get

$$c(t) = (\cos t, \sin t). \quad (2.12)$$

This integral curve is a circle centered at the origin with radius 1, and it is defined for all $t \in \mathbb{R}$. If, instead, we had chosen $p_0 = (x_0, y_0) \in \mathbb{R}^2$, then the integral curve of X starting at p_0 would be given by

$$c(t) = (x_0 \cos t - y_0 \sin t, x_0 \sin t + y_0 \cos t), \quad (2.13)$$

which is a circle centered at the origin with radius $\sqrt{x_0^2 + y_0^2}$, and it is also defined for all $t \in \mathbb{R}$. Moreover, one can verify that $c_s(c_t) = c_{s+t}$ for all $s, t \in \mathbb{R}$, where $c_t(p_0)$ denotes the integral curve of X starting at p_0 evaluated at time t . c_t is a diffeomorphism of $\mathbb{R}^2$ for each $t \in \mathbb{R}$, and the map

$$\mathbb{R} \rightarrow \text{Diff}(\mathbb{R}^2), \quad t \mapsto c_t \quad (2.14)$$

is a smooth group homomorphism from $(\mathbb{R}, +)$ to the group of diffeomorphisms of $\mathbb{R}^2$, where the group operation on $\text{Diff}(\mathbb{R}^2)$ is composition of maps.

From the above, any group homomorphism from $(\mathbb{R}, +)$ to the group of diffeomorphisms of a smooth manifold M is called a one-parameter group of diffeomorphisms of M .

Corollary 2.21. *Let X be a smooth vector field on a smooth manifold M , with $p \in M$. Then there exists an integral curve $c : (a, b) \rightarrow M$ of X such that $0 \in (a, b)$ with $c(0) = p$, and this integral curve is unique.*

Theorem 2.22. *Let $V \subseteq \mathbb{R}^n$ be open, and $f : V \rightarrow \mathbb{R}^n$ be a smooth map. Then for each point $p_0 \in V$, there exists a neighbourhood W of p_0 in V , an $\epsilon > 0$, and a smooth function $y : (-\epsilon, \epsilon) \times W \rightarrow V$ such that*

$$\frac{\partial y}{\partial t}(t, q) = f(y(t, q)), \quad y(0, q) = q \quad (2.15)$$

for all $(t, q) \in (-\epsilon, \epsilon) \times W$.

Corollary 2.23. *Let X be a smooth vector field on a smooth manifold M , and let U be a chart around $p \in M$. Then there exists a neighbourhood W of p in U , an $\epsilon > 0$, and a smooth map $F : (-\epsilon, \epsilon) \times W \rightarrow U$ such that for all $q \in W$, $F(t, q)$ is an integral curve of X with initial point q ($F(0, q) = q$).*

In the above, we use the notation $F(t, q) = F_t(q)$. Suppose $s, t \in (-\epsilon, \epsilon)$, and $q \in W$, and that $F_t(F_s(q))$ and $F_{t+s}(q)$ are both defined. Then we have $F_t(F_s(q)) = F_{t+s}(q)$. This is because both sides are integral curves of X starting at q at time 0, and by uniqueness of integral curves, they must coincide. F , above, is termed as a *local flow* generated by the vector field X . $F_t(q)$ is then a flow line of q of the local flow F . If F can be defined as $F : \mathbb{R} \times M \rightarrow M$, then F is called a *global flow* generated by the vector field X . X is called a *complete vector field* if its flow is global.

Definition 2.24. In general, a *local flow* about a point $p \in U \subseteq M$ is a smooth map $F : (-\epsilon, \epsilon) \times W \rightarrow U$, where W is a neighbourhood of p in U and $\epsilon > 0$, such that

1. $F_0(q) = q$ for all $q \in W$, and
2. $F_t(F_s(q)) = F_{t+s}(q)$ whenever both sides are defined.

2.4 Lie Algebras and Lie Groups

September 23rd.

Recall that a vector field X on a smooth manifold M was smooth if and only if for any smooth function $f \in C^\infty(M)$, the function $Xf : M \rightarrow \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. That is, the map Xf itself is a smooth map on M ; if $f \in C_p^\infty$, then $Xf \in C_p^\infty$ as well. If we have another smooth vector field Y on M , then we can define a new map $[X, Y](\cdot)(p) : C^\infty(M) \rightarrow \mathbb{R}$ by

$$f \mapsto X_p(Yf) - Y_p(Xf). \quad (2.16)$$

This map is $\mathbb{R}$ -linear, and it is also a derivation on $C^\infty(M)$; that is, for any $f, g \in C^\infty(M)$, we have

$$\begin{aligned} [X, Y](fg)(p) &= X_p(Y(fg)) - Y_p(X(fg)) = X_p(fYg + gYf) - Y_p(fXg + gXf) \\ &= f(p)X_p(Yg) + g(p)X_p(Yf) - f(p)Y_p(Xg) - g(p)Y_p(Xf) \\ &= f(p)(X_p(Yg) - Y_p(Xg)) + g(p)(X_p(Yf) - Y_p(Xf)) \\ &= f[X, Y](g)(p) + g[X, Y](f)(p). \end{aligned} \quad (2.17)$$

Thus, $[X, Y](\cdot)(p)$ is a derivation on $C^\infty(M)$, and so there exists a unique tangent vector $[X, Y]_p \in T_pM$ such that $[X, Y]_p(f) = [X, Y](f)(p)$ for all $f \in C^\infty(M)$. This defines a new vector field $[X, Y]$ on M , defined as

$$XY - YX = [X, Y] : M \rightarrow TM, \quad p \mapsto [X, Y]_p. \quad (2.18)$$

This is called the Lie bracket of the vector fields X and Y , and is smooth when X and Y are smooth vector fields on M . $\mathfrak{X}(U)$ then, the set of all smooth vector fields on an open set $U \subseteq M$, is termed a Lie algebra.

Definition 2.25. Let k be a field and V be a vector space over k . V is termed a *Lie algebra* over k when equipped with an operation $[\cdot, \cdot] : V \times V \rightarrow V$ called the Lie bracket, such that the following conditions hold:

1. bilinearity: $[av_1 + bv_2, w] = a[v_1, w] + b[v_2, w]$ and $[v, aw_1 + bw_2] = a[v, w_1] + b[v, w_2]$ for all $v, v_1, v_2, w, w_1, w_2 \in V$ and $a, b \in k$,
2. anti-commutativity: $[v, w] = -[w, v]$ for all $v, w \in V$, and
3. Jacobi identity: $[v, [w, u]] + [w, [u, v]] + [u, [v, w]] = 0$ for all $v, w, u \in V$.

Verify that the set of smooth vector fields $\mathfrak{X}(U)$ on an open set $U \subseteq M$ forms a Lie algebra over $\mathbb{R}$ with the Lie bracket $[\cdot, \cdot] : \mathfrak{X}(U) \times \mathfrak{X}(U) \rightarrow \mathfrak{X}(U)$ defined above as $[X, Y] = XY - YX$ for all $X, Y \in \mathfrak{X}(U)$.

Any vector V over the field k can be viewed as a Lie algebra over k with the trivial Lie bracket $[v, w] = 0$ for all $v, w \in V$. This is also the only abelian Lie algebra over k ; that is, a Lie algebra V over k is abelian if and only if $[v, w] = 0$ for all $v, w \in V$. The vector space $M_n(k)$ of all $n \times n$ matrices over a field k is a Lie algebra over k with the Lie bracket defined as $[A, B] = AB - BA$ for all $A, B \in M_n(k)$.

Definition 2.26. Let $F : N \rightarrow M$ be a smooth map between smooth manifolds, and let $p \in N$ with the induced linear map $F_{*,p} : T_pN \rightarrow T_{F(p)}M$. If X is a vector field on N and Y is a vector field on M , we say that X and Y are *F-related* if $F_{*,p}(X_p) = Y_{F(p)}$ for all $p \in N$. Y is called the pushforward of X by F .

Lemma 2.27. With notation as above, X and Y are *F-related* if and only if for any smooth function $g \in C^\infty(M)$, we have $X(g \circ F) = (Yg) \circ F$.

Proof. Suppose X and Y are *F-related*. Then for any smooth function $g \in C^\infty(M)$, we have

$$X(g \circ F) = F_{*,p}(X_p)(g) = Y_{F(p)}(g) = (Yg) \circ F.$$

Conversely, suppose that for any smooth function $g \in C^\infty(M)$, we have $X(g \circ F) = (Yg) \circ F$. Then for any smooth function $g \in C^\infty(M)$, we have

$$F_{*,p}(X_p)(g) = X(g \circ F) = (Yg) \circ F = Y_{F(p)}(g).$$

Thus, $F_{*,p}(X_p) = Y_{F(p)}$ for all $p \in N$, and so X and Y are F -related. ■

Lemma 2.28. *With notation as above, if X_1, X_2 are vector fields on N and Y_1, Y_2 are vector fields on M such that X_i and Y_i are F -related for $i = 1, 2$, then the Lie brackets $[X_1, X_2]$ and $[Y_1, Y_2]$ are also F -related.*

Proof. Suppose X_1, X_2 are vector fields on N and Y_1, Y_2 are vector fields on M such that X_i and Y_i are F -related for $i = 1, 2$. Then for any smooth function $g \in C^\infty(M)$, we have

$$\begin{aligned} [X_1, X_2](g \circ F) &= X_1(X_2(g \circ F)) - X_2(X_1(g \circ F)) \\ &= X_1((Y_2g) \circ F) - X_2((Y_1g) \circ F) \\ &= (Y_1(Y_2g)) \circ F - (Y_2(Y_1g)) \circ F \\ &= ([Y_1, Y_2]g) \circ F. \end{aligned}$$

Thus, by the previous lemma, we have that $[X_1, X_2]$ and $[Y_1, Y_2]$ are F -related. ■

September 25th.

Definition 2.29. A Lie group G is both a group and a smooth manifold such that the maps

$$G \times G \rightarrow G, \quad (g, h) \mapsto gh, \quad G \rightarrow G, \quad g \mapsto g^{-1}. \quad (2.19)$$

are smooth maps.

For example, both $GL_n(\mathbb{R})$ and $SL_n(\mathbb{R})$ are Lie groups. Similarly, $O(n)$ is also a Lie group. $(\mathbb{R}^n, +)$ is also a Lie group. If G and H are both Lie groups, then so is $G \times H$.

Definition 2.30. If G and H are Lie groups, a map of Lie groups $f : G \rightarrow H$ is a map that is both a homomorphism of groups and a smooth map between smooth manifolds.

The determinant $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$ is a map of Lie groups.

Definition 2.31. Let G be a Lie group. A subset $H \subseteq G$ is termed a Lie subgroup if H is a subgroup and an immersed manifold of G , and the product and inverse operations restricted to H remain smooth.

Example 2.32. We look at the space $S^1 \times S^1 = \mathbb{R}^2/\mathbb{Z}^2$. Let f be the natural quotient map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$. Look at the map $g_\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ defined as $t \mapsto (t, \alpha t)$ for some $\alpha \in \mathbb{R}$. Let $L_\alpha = g_\alpha(\mathbb{R}) = \{(t, \alpha t) \mid t \in \mathbb{R}\}$ be the image line, and now set $H_\alpha = f(L_\alpha) = f \circ g_\alpha(\mathbb{R})$. We claim that $f \circ g_\alpha$ is a one-to-one immersion, and further that H_α is a closed curve if and only if α is rational. The map f may be explicitly written as $(s, t) \mapsto (e^{2\pi i s}, e^{2\pi i t})$. Note that it is enough to prove that $f^{-1}(H_\alpha) = \{(t + n, \alpha t + m) \mid t \in \mathbb{R}, n, m \in \mathbb{Z}\}$ is dense in $\mathbb{R}^2$ if and only if α is irrational. If $\alpha = \frac{p}{q}$ is rational, then $f^{-1}(H_\alpha) = \{(t + n, \frac{p}{q}t + m) \mid t \in \mathbb{R}, n, m \in \mathbb{Z}\}$. Then for any $t \in \mathbb{R}$, we have $(t + n, \frac{p}{q}t + m) = (t + n, \frac{p}{q}(t + qn) + (m - pn))$, and so $f^{-1}(H_\alpha) = \{(t + n, \frac{p}{q}t + m) \mid t \in [0, q), n, m \in \mathbb{Z}\}$. This is a union of q lines in $\mathbb{R}^2$, and so it cannot be dense in $\mathbb{R}^2$. Conversely, if α is irrational, then for any $(x, y) \in \mathbb{R}^2$ and $\epsilon > 0$, we can find integers n, m such that $|x - n| < \epsilon$ and $|y - m - \alpha n| < \epsilon$. This shows that $f^{-1}(H_\alpha)$ is dense in $\mathbb{R}^2$ if and only if α is irrational.

2.4.1 Matrix Lie Groups and Lie Algebras

September 28th.

Definition 2.33. For a matrix $A \in M_n(\mathbb{R})$, we define the exponential map $\exp : M_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ as

$$\exp(A) = e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \quad (2.20)$$

To show that the exponential map is well-defined and lands in $GL_n(\mathbb{R})$, we need to show that the series converges and that the resulting matrix is invertible. For $X \in M_n(\mathbb{R})$, define the norm $\|X\| = \sqrt{\sum_{i,j=1}^n x_{ij}^2}$; this norm is also sub-multiplicative, meaning that $\|XY\| \leq \|X\| \|Y\|$ for all $X, Y \in M_n(\mathbb{R})$. Then, looking at the partial sums of the series, we have

$$\left\| \sum_{k=0}^N \frac{A^k}{k!} \right\| \leq \sum_{k=0}^N \frac{\|A\|^k}{k!} \leq \|I\| + \sum_{k=1}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|} + \sqrt{n} - 1 \quad (2.21)$$

which implies that the series converges absolutely, and hence converges. We also state a few properties of the trace map $\text{tr} : M_n(\mathbb{R}) \rightarrow \mathbb{R}$ defined as $\text{tr}(A) = \sum_{i=1}^n a_{ii}$ for $A = (a_{ij}) \in M_n(\mathbb{R})$; in particular, the following properties hold:

1. $\text{tr}(AB) = \text{tr}(BA)$ for all $A, B \in M_n(\mathbb{R})$.
2. If $A \in GL_n(\mathbb{R})$, then $\text{tr}(AXA^{-1}) = \text{tr}(X)$ for all $X \in M_n(\mathbb{R})$.
3. For $A \in M_n(\mathbb{R})$, $\text{tr}(A) = \sum_{i=1}^n \lambda_i$, where $\lambda_1, \dots, \lambda_n$ are the (complex) eigenvalues of A .
4. For all $A \in M_n(\mathbb{R})$, there exists $B \in GL_n(\mathbb{C})$ such that BAB^{-1} is upper triangular, and has the eigenvalues of A on the diagonal. In particular, $\det(A) = \prod_{i=1}^n \lambda_i$, where $\lambda_1, \dots, \lambda_n$ are the (complex) eigenvalues of A .

Proposition 2.34. For $A \in M_n(\mathbb{R})$, $\det e^A = e^{\text{tr}(A)}$.

Proof. Without the loss of generality, we may assume that A is upper triangular, with eigenvalues $\lambda_1, \dots, \lambda_n$ on the diagonal (if not, we can find $B \in GL_n(\mathbb{C})$ such that BAB^{-1} is upper triangular, and $\det e^A = \det e^{BAB^{-1}}$ and $\text{tr}(A) = \text{tr}(BAB^{-1})$). Then e^A is also upper triangular, with eigenvalues $e^{\lambda_1}, \dots, e^{\lambda_n}$ on the diagonal. Therefore, we have

$$\det e^A = \prod_{i=1}^n e^{\lambda_i} = e^{\sum_{i=1}^n \lambda_i} = e^{\text{tr}(A)}. \quad (2.22)$$

■

In particular, if $A \in M_n(\mathbb{R})$, then $e^A \in GL_n(\mathbb{R})$, and so the exponential map $\exp : M_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is well-defined. We use this exponential map with smooth curves. Define $c : \mathbb{R} \rightarrow GL_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$ as $c(t) = e^{tA}$ for some $A \in M_n(\mathbb{R})$. Then c is a smooth curve in $GL_n(\mathbb{R})$, with $c(0) = I$ and $c'(0) = A$. In fact, the derivative is given as

$$c'(t) = \frac{d}{dt} e^{tA} = \frac{d}{dt} \left(\sum_{k=0}^{\infty} \frac{t^k A^k}{k!} \right) = \sum_{k=1}^{\infty} \frac{k t^{k-1} A^k}{k!} = \sum_{k=1}^{\infty} \frac{t^{k-1} A^k}{(k-1)!} = A \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = A e^{tA}. \quad (2.23)$$

Now, let G be a Lie group, and let $g \in G$. Define the left translation map $\ell_g : G \rightarrow G$ as $\ell_g(h) = gh$ for all $h \in G$. Then ℓ_g is a diffeomorphism of G , with inverse $\ell_{g^{-1}}$. Let $G = GL_n(\mathbb{R})$. Then the induced linear map at the identity is

$$(\ell_g)_{*,I} : T_I(GL_n(\mathbb{R})) \rightarrow T_g(GL_n(\mathbb{R})) \quad (2.24)$$

With $c : \mathbb{R} \rightarrow GL_n(\mathbb{R})$ a smooth curve with $c(0) = I$ and $c'(0) = A$, we have

$$(\ell_g)_{*,I}(A) = (\ell_g \circ c)'(0) = (gc)'(0) = gc'(0) = gA. \quad (2.25)$$

The exact same map is induced. If we instead look at the map $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$, then the induced linear map at the identity is

$$(\det)_{*,I} : T_I(GL_n(\mathbb{R})) \rightarrow T_1(\mathbb{R}^*) \quad (2.26)$$

and repeating the above, we have

$$(\det)_{*,I}(A) = (\det \circ c)'(0) = \left. \frac{d}{dt} \det(c(t)) \right|_{t=0} = \left. \frac{d}{dt} \det(e^{tA}) \right|_{t=0} = \left. \frac{d}{dt} e^{\text{tr}(tA)} \right|_{t=0} = \text{tr}(A). \quad (2.27)$$

That is, the induced linear map $(\det)_{*,I} : T_I(GL_n(\mathbb{R})) \rightarrow T_1(\mathbb{R}^*)$ is given as $A \mapsto \text{tr}(A)$. Now, pre-add the inclusion map $i : SL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$, and consider the composition $\det \circ i : SL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$. Then the induced linear map at the identity is

$$(\det \circ i)_{*,I} : T_I(SL_n(\mathbb{R})) \rightarrow T_1(\mathbb{R}^*). \quad (2.28)$$

But with the chain rule for induced linear maps, we have

$$(\det \circ i)_{*,I} = (\det)_{*,I} \circ (i)_{*,I} : T_I(SL_n(\mathbb{R})) \rightarrow T_1(\mathbb{R}^*), \quad A \mapsto \text{tr}(A) = 0. \quad (2.29)$$

That is, the induced linear map $(\det \circ i)_{*,I} : T_I(SL_n(\mathbb{R})) \rightarrow T_1(\mathbb{R}^*)$ is the zero map. This shows that $T_I(SL_n(\mathbb{R})) = \ker((\det \circ i)_{*,I}) = \{A \in M_n(\mathbb{R}) \mid \text{tr}(A) = 0\}$, which is a vector space of dimension $n^2 - 1$. From here, we define

$$\mathfrak{sl}_n(\mathbb{R}) := T_I(SL_n(\mathbb{R})) = \{A \in M_n(\mathbb{R}) \mid \text{tr}(A) = 0\}, \quad (2.30)$$

the space of traceless matrices. Moving on, we look at the Lie algebra of $O(n)$. Let $G = O(n) \subseteq GL_n(\mathbb{R})$ and consider the map $f : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ defined as $f(A) = A^T A$. Then $O(n) = f^{-1}(I)$, and so the induced linear map at the identity is

$$f_{*,I} : T_I(GL_n(\mathbb{R})) \rightarrow T_I(GL_n(\mathbb{R})). \quad (2.31)$$

Pre-composing the smooth curve $c : \mathbb{R} \rightarrow GL_n(\mathbb{R})$ defined as $c(t) = e^{tA}$, we have

$$f_{*,I}(A) = (f \circ c)'(0) = (f(e^{tA}))'(0) = (e^{tA^T} e^{tA})'(0) = (e^{t(A^T + A)})'(0). \quad (2.32)$$

Thus, we have $T_I(O(n)) = \ker(f_{*,I}) = \{A \in M_n(\mathbb{R}) \mid A^T + A = 0\}$, which is a vector space of dimension $\frac{n(n-1)}{2}$. From here, we define

$$\mathfrak{so}_n(\mathbb{R}) := T_I(O(n)) = \{A \in M_n(\mathbb{R}) \mid A^T + A = 0\}, \quad (2.33)$$

the space of skew-symmetric matrices. We also denote $\mathfrak{gl}_n(\mathbb{R}) := T_I(GL_n(\mathbb{R})) = M_n(\mathbb{R})$, the space of all $n \times n$ matrices. These structures are all Lie algebras (subalgebras of $\mathfrak{gl}_n(\mathbb{R})$) with the Lie bracket defined as $[A, B] = AB - BA$ for all $A, B \in \mathfrak{gl}_n(\mathbb{R})$. In particular, we have the inclusions

$$\mathfrak{so}_n(\mathbb{R}) \subseteq \mathfrak{sl}_n(\mathbb{R}) \subseteq \mathfrak{gl}_n(\mathbb{R}). \quad (2.34)$$

It is left as an exercise to show that the Lie bracket is closed on $\mathfrak{so}_n(\mathbb{R})$ and $\mathfrak{sl}_n(\mathbb{R})$.

One may show that if $AB = BA$, then $e^{A+B} = e^A e^B$, and $(e^A)^{-1} = e^{-A}$.

Definition 2.35. Let G be a Lie group, and X be a (not necessarily smooth) vector field on G . X , then, is called a *left-invariant vector field* on G if $(\ell_g)_* X = X$ for all $g \in G$, or

$$(\ell_g)_* X_h = X_{\ell_g(h)} = X_{gh} \quad (2.35)$$

for any $g, h \in G$. That is, X is left-invariant if it is invariant under the pushforward of left translation by any element of G .

If we pick the identity $e \in G$, the above property of the vector field X boils down to the property that $X_g = (\ell_g)_* X_e$ for all $g \in G$. That is, a left-invariant vector field is completely determined by its value at the identity.

Proposition 2.36. *Any left-invariant vector field on a Lie group G is smooth.*

Proof. Let X be a left-invariant vector field on a Lie group G . Note that it is enough to prove that if $f \in C^\infty(G)$, then $Xf \in C^\infty(G)$. Let $g \in G$, and let $c : \mathbb{R} \rightarrow G$ be a smooth curve with $c(0) = e$ and $c'(0) = X_e$. Then, we have

$$(Xf)(g) = X_g(f) = (\ell_g)_* X_e(f) = X_e(f \circ \ell_g) = \left. \frac{d}{dt} (f \circ \ell_g \circ c(t)) \right|_{t=0}. \quad (2.36)$$

The map $(g, t) \mapsto f \circ \ell_g \circ c(t)$ is smooth, and so the map $g \mapsto (Xf)(g)$ is smooth as well. Thus, $Xf \in C^\infty(G)$, and so X is a smooth vector field on G . ■

Suppose X and Y are two left-invariant vector fields on a Lie group G . That means, X and Y are ℓ_g -related for all $g \in G$. Thus, by the result above, $[X, Y]$ must be ℓ_g -related, which tells us $[X, Y]$ is a left-invariant vector field on G . Thus, we have, from before,

$$\mathfrak{X}(G) = \{X : G \rightarrow TG \mid X \text{ is smooth}\}, \quad (2.37)$$

the set of all smooth vector fields on G along with the Lie bracket $[X, Y] = XY - YX$ is a Lie algebra, and

$$L(G) := \{X : G \rightarrow TG \mid X \text{ is left-invariant}\} \subseteq \mathfrak{X}(G) \quad (2.38)$$

is a Lie subalgebra of $\mathfrak{X}(G)$. That is, there is a bijection between left-invariant vector fields on G and the tangent space at the identity $T_e G$. In particular, we have

$$L(G) \cong T_e G, \quad X \mapsto X_e \quad (2.39)$$

where the inverse map is given as $v \mapsto X_v$, where X_v is the unique left-invariant vector field on G such that $X_{v,e} = v$. Thus, we have that $L(G)$ is a finite-dimensional Lie algebra over $\mathbb{R}$, with dimension $\dim L(G) = \dim T_e G = \dim G$. This bijection is as $\mathbb{R}$ -linear spaces, and the Lie bracket on $L(G)$ is given as $[X_v, X_w] = X_{[v,w]}$ for all $v, w \in T_e G$. The Lie algebra $L(G)$ is called the Lie algebra of the Lie group G . In particular, we have that $L(GL_n(\mathbb{R})) = \mathfrak{gl}_n(\mathbb{R})$, $L(SL_n(\mathbb{R})) = \mathfrak{sl}_n(\mathbb{R})$, and $L(O(n)) = \mathfrak{o}_n(\mathbb{R})$.

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